MILITARY SPECIFICATION

COMMUNICATIONS SETS, UHF
AN/WSC-3(V) SATELLITE AND LOS VERSIONS

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for the design, manufacture, and testing of an ultra high frequency (UHF) Satellite Communications (SATCOM) Set, including an internal modem, for shipboard installation hereinafter referred to as the equipment. The equipment is capable of transmitting and receiving information over the frequency range of 225.000 megahertz (MHz) to 399.975 MHz. The transmitter section of the equipment has a minimum carrier power output of 100 watts (W) for frequency shift keying (FSK) and phase shift keying (PSK) modulation and 30 W for amplitude modulation (AM) in a continuous duty cycle under the environmental conditions specified herein.

1.2 Line-of-sight (LOS) version. This specification also covers an amplitude and frequency modulated LOS version. The LOS version provides plain and secure AM and frequency modulation (FM) capability using TSEC/KY-8 and TSEC/KY-58 Secure Voice equipment for LOS communications only.

1.3 Nomenclature. TABLE I specifies the AN/WSC-3(V) nomenclature. All versions except the AN/WSC-3(V)4, (V)10, and (V)11 are specified herein.

TABLE I. AN/WSC-3(V) nomenclature.

Nomenclature	Equipment
AN/WSC-3(V)1	Same as basic AN/WSC-3 SATCOM set but with FM/VINSON, LINK 11, and single audio system (SAS) chassis wiring but no VINSON, LINK 11, or SAS modules
AN/WSC-3(V)2	(V)1 SATCOM radio with FM/VINSON and LINK 11 modules, standard ship input/output (I/O) interface module
AN/WSC-3(V)3	(V)2 SATCOM radio, but with an SAS ship I/O interface module
AN/WSC-3(V)4	AM-LOS radio (not described by this specification)
AN/WSC-3(V)5	AM-LOS radio with FM/VINSON, LINK 11, and SAS chassis wiring, but no VINSON, LINK 11, or SAS modules
AN/WSC-3(V)6	AM/FM LOS radio; (V)5 radio with VINSON and LINK 11 modules; standard ship I/O interface module
AN/WSC-3(V)7	(V)6 LOS radio but with a SAS ship I/O interface module
AN/WSC-3(V)8	(V)7 radio but with Coast Guard serial control

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5820

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TABLE I. AN/WSC-3(V) nomenclature. (continued)

Nomenclature	Equipment
AN/WSC-3(V)9	SATCOM radio similar to (V)3 except modified for US AIR FORCE to tune in 5-kHz increments
AN/WSC-3(V)10	(V)6 LOS radio modified for HAVE QUICK
AN/WSC-3(V)11	(V)7 LOS radio modified for HAVE QUICK
AN/WSC-3(V)12	(V)3 SATCOM radio modified for installation in TRIDENT integrated radio room

1.4 Commonality of AN/WSC-3(V) versions. The LOS versions of the AN/WSC-3(V) are identical to the SATCOM versions both in electrical and mechanical composition, except that modules associated with SATCOM are removed, and jumper modules are substituted which provide necessary chassis circuit continuity and I/O interface characteristics, including built-in-test equipment (BITE) function capability. Modules associated with SATCOM are not delivered with the LOS version unless authorized by the procuring activity (see 6.2.1). The LOS radio front panel controls are identical to the SATCOM versions, but small identification plates are panel-mounted. The identification plate will bear a statement that the radio is configured for LOS operation only. Black dots are placed next to the numbers on the TEST SELECT switch associated with LOS BITE functions. It is possible to restore SATCOM capability to an LOS radio by removing the jumper modules, inserting SATCOM modules, and removing the LOS identification plate. No other changes or modifications to the equipment are required.

1.5 Functional configuration matrix. The AN/WSC-3(V) versions exhibit the modulation modes specified in TABLE II.

TABLE II. Modulation modes for AN/WSC-3(V) versions.

Communication mode	AN/WSC-3(V) versions										
	(V)1	(V)2	(V)3	(V)5	(V)6	(V)7	(V)8	(V)9	(V)10	(V)11	(V)12
AM Plain	X	X	X	X	X	X	X	X	X	X	X
AM Secure	X	X	X	X	X	X	X	X	X	X	X
FM Plain	X	X	X		X	X	X	X	X	X	X
FM Secure		X	X		X	X	X	X	X	X	X
FSK	X	X	X					X			X
PSK	X	X	X					X			X
EXIT Modem	X	X	X					X			X
LINK 11		X	X		X	X	X	X	X	X	X
AJ									X	X	

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbook. The following specifications, standards, and handbook form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standard (DoDISS) and supplement thereto, cited in the solicitation.

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SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High Impact), Shipboard Machinery, Equipment And Systems, Requirements For
MIL-C-7438	Core Material Aluminum For Sandwich Construction
MIL-M-7793	Meter, Time Totalizing
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts), Packaging Of
MIL-C-22954	Control, Radio Set C-1138( )/UR
MIL-M-38510	Microcircuits, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirement For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-188	Military Communication System Technical Standards
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-740	Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)

HANDBOOK

MILITARY

MIL-HDBK-175	Microelectronic Device Data Handbook

2.1.2 Other Government documents, drawing, and publications. The following other government documents, drawing, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

MILITARY

RE-F-2688036	H-169/U Handset

PUBLICATIONS

NATIONAL SECURITY AGENCY (NSA)

CSEEB-6	Engineering Bulletin TSEC/KY-8
CSEEB-32	Engineering Bulletin TSEC/KY-58
CSESD-14	System Document TSEC/KY-58

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DEFENSE NUCLEAR AGENCY (DNA)

DNA 1420H	TREE (Transient Radiation Effect Of Electronics) Handbook
DNA 2114H	EMP (Electromagnetic Pulse) Handbook

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536D	Procedures And Requirements For Preparation And Soldering Of Electrical Connections

NAVAL SEA SYSTEMS COMMAND

NAVSEA 0967-LP-143-4010	Teletypewriter-Transmitter Control C-1004/SG

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

NAVELEX PME 116 11019/SAL Ser. 00684 of 5 March 1976	EMP and TREE Hardening Specification, Nuclear Environment Statement

BUREAU OF NAVAL PERSONNEL

NAVPERS 18068	Manual Of Qualifications For Advancement

(Copies of specifications, standards, handbook, drawing, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77	Racks, Panels, And Associated Equipment

(Request for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The equipment procured in accordance with the requirements of this specification shall be tactical UHF communications sets for shipboard and submarine installation, for SATCOM and LOS use. The equipment shall operate in a simplex mode and shall be capable of transmitting and receiving voice, teletype, and digital data signals via Department of Defense (DoD) tactical UHF satellite or general communications via LOS. The equipment shall use, to the maximum extent practical, microcircuit techniques and solid state technology. The equipment, including parts, materials, and processes, shall comply with the requirements of MIL-E-16400, to the extent specified herein. The design objectives as specified in MIL-E-16400 for reliability, simplicity, and ease of maintenance shall be considered during all design phases of the equipment.

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3.1.1 Interface requirements. The radios shall interface with the equipment specified herein. For satellite communications, OE-82B/WSC-1(V), OE-176/BRQ, and AN/BRA-34 antenna groups shall be compatible, as well as OK-326A/WSC-3 and OK-327/WSC-3 equipment racks. For LOS communications, AS-390/SRC, AS-1018/URC, and AS-1735/SRC antenna groups, AN/SRA-33 and OA-9123/SRC multicouplers, MT-6068/WSC-3(V), and MT-6069/WSC-3(V) equipment racks shall be compatible. During either mode of communication, the equipments shall interface with TSEC/KY-8 and TSEC/KY-58 security devices, C-1138/UR radio set control, H-169/U handset, and AN/USQ-36, AN/USQ-74, and data terminal sets. TA-840A/U, TA-970/U, and TA-980/U plain and secure voice phone sets shall also interface with all equipment. The SA-2254/UR SAS adapter box is used with all radio versions except the (V)3 and (V)7 versions.

3.1.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Shock and vibration requirements. The equipment shall conform to the shock requirements for Grade A, Class I, lightweight, Type A of MIL-S-901. The vibration requirements for the equipment shall be Type I of MIL-STD-167-1. The equipment shall comply with the shock and vibration requirements without internal or external resilient mountings.

3.1.2.1 Inclination. The equipment shall be in accordance with the inclination requirements of MIL-E-16400. The inclination angle shall be 50 degrees.

3.1.3 Temperature and humidity requirements. The equipment shall comply with the Class 4 temperature requirements of MIL-E-16400. The humidity requirements from 0 percent to 95 percent relative humidity, shall be in accordance with MIL-E-16400.

3.1.4 Airborne and structureborne noise requirements. The airborne noise requirements for the equipment shall be as specified in TABLE III. The octave band sound pressure level (L_p), expressed in decibels (dB) with respect to a reference level of 0.0002 microbar and when the measurement is taken at 0.91 meters (m) (3 feet (ft)) from all surfaces measured along the equipment centerlines, shall not exceed that specified in TABLE III. The speech-interference-level (SIL), average sound pressure level, shall not exceed 60 dB in the 600 Hertz (Hz) to 1200 Hz, 1200 Hz to 2400 Hz, and 2400 Hz to 4800 Hz octave bands. The equipment shall comply with the structure-borne noise requirements specified in MIL-STD-740.

TABLE III. Airborne noise.

Octave band (Hz)	Lp (dB/0.0002 microbar)
37.5 to 75	79
75 to 150	73
150 to 300	68
300 to 600	63
600 to 1200	62
1200 to 2400	60
2400 to 4800	59
4800 to 9600	57

3.1.5 Design and construction. The design and construction of the equipment shall conform to the requirements of MIL-E-16400. The equipment shall be designed and fabricated using solid-state electronic circuitry throughout without the use of vacuum tubes and electromechanical tuning for frequency selection.

3.1.5.1 Parts and processes. Parts and processes shall be as specified in 3.1.5.1.1 and 3.1.5.1.2.

3.1.5.1.1 Parts. Parts shall be as specified in 3.1.5.1.1.1 through 3.1.5.1.1.5.

3.1.5.1.1.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

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3.1.5.1.1.2 Mating connectors. A complete set of mating halves of connectors, backshells, and protective caps shall be provided. Except for coaxial type they shall be solderless (crimp-on) type whenever possible.

3.1.5.1.1.3 Microelectronic modular assemblies. All microelectronic assemblies and devices employed in the design of the equipment shall comply with Class B requirements of MIL-M-38510. MIL-HDBK-175 shall be used as guidance in the selection and application of all microelectronic devices used in the equipment.

3.1.5.1.1.4 Module sockets. All module sockets shall be readily accessible for easy removal and replacement.

3.1.5.1.1.5 Printed wiring boards. Printed wiring boards shall conform to MIL-STD-454, Requirement 17.

3.1.5.1.2 Structural welding. Compliance with the Structural welding paragraph of MIL-E-16400 is not required provided that such noncompliance does not compromise the shock, vibration, and workmanship requirements specified in 3.1.2 and 3.8.

3.1.6 Circuit test points. Circuit test points shall be provided which comply with the criteria specified in MIL-STD-415. The types of tests specified in a through h shall be considered for determining test points:

a. Signal quality
b. Circuit performance
c. Calibration
d. System alignment
e. Installation and checkout
f. Planned maintenance system tests
g. Line replaceable unit (LRU) fault location
h. Shop replaceable part (SRP) fault isolation

Test points for calibration shall be provided external to the unit under test, except where it would affect the calibration. Test points shall be provided with the proper interface fittings and matching circuits to connect the selected support test equipment.

3.1.7 Equipment cooling. If forced air cooling of the equipment is required, the air intake shall be from the front and shall be exhausted from the rear of the equipment. All air intakes shall be covered by a suitable air filter which can be removed without the necessity for removal of adjacent assemblies. The air filters shall be a permanent type which can be cleaned with soap and water. The use of direct current (DC) blower motors in the equipment is prohibited.

3.1.7.1 High temperature protection. The equipment shall employ fail-safe devices which protect the equipment from damage in the event of loss of cooling air or in the event of excessively high ambient temperatures. The fail-safe devices shall also provide protection while the equipment is in the standby mode.

3.1.8 Dimensions. The front panel of the equipment shall be for a standard 482.6 millimeters (mm) (19-inch (in.)) electronic equipment rack mount installation in accordance with EIA RS-310-C-77 except as specified herein. The overall equipment height shall not exceed 311.150 mm (12.25 in.). The overall equipment depth shall not exceed 587.5 mm (23.13 in.). The overall distance from the rack mounting surface of the equipment to the rear of the equipment, excluding mating connectors and plugs, shall not exceed 524 mm (20.63 in.). The equipment shall not protrude more than 63.5 mm (2.5 in.) in front of the equipment rack mounting surface. The total weight of the equipment shall not exceed 70.29 kilograms (kg) (155 pounds (lbs)).

3.1.9 Degree of enclosure. The cabinet for the equipment shall be splash-proof construction in accordance with MIL-STD-108.

3.1.10 Altitude, mean-sea-level (MSL). The equipment shall be capable of operating at 1.524 kilometers (km) (5000 ft) MSL under the service conditions specified in 3.1.19. The equipment shall survive without any damage at 15.24 km (50,000 ft) MSL while not operating.

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3.1.11 Electrical overload protection. Electrical overload protection shall be provided in the equipment in accordance with the requirements of the Electrical overload protection paragraph of MIL-E-16400. All parts and circuits in the equipment which are likely to carry an overload due to any failure or improper adjustments shall have protection devices, or shall be properly rated to carry an overload without permanent damage to the equipment.

3.1.11.1 Electrical fuses. All fuses shall be located on the front panel of the equipment. Blown fuse indicators shall be used in the equipment. The spare fuses shall be located accessibly inside the equipment.

3.1.12 Handles. Handles shall be provided on the front panel of the equipment to ensure safe and easy removal of the equipment from the mounting. The handles shall be at least 12.7 mm (0.50 in.) in width with no sharp edges. Each drawer which is removable from the cabinet shall have suitable handles or bails. Handles shall not protrude more than 50.8 mm (2 in.) from the front panel surface. All controls on the front panel shall not protrude beyond the handles.

3.1.13 Locking devices. Locking devices are required on all critical internal adjustment controls in order to prevent the need for readjustment of the controls after sustained periods of shock and vibration.

3.1.14 Equipment slide and tilt mechanisms. The equipment shall have slide mechanisms which shall be of suitably rugged construction to support the drawer in the withdrawn position. The slide mechanisms shall have a locking device to secure the drawer in the withdrawn position. A tilt mechanism shall be provided on the drawer slides to permit ±90-degree tilt and locking. The slide mechanism shall slide with a minimum of sliding friction without a tendency to bind. The slide and tilt mechanisms shall be constructed of corrosion-resistant material, such as corrosion-resistant steel. Guide support pins shall be provided between the rear of each chassis and the enclosure to prevent possible damage, due to shock, to the chassis slides.

3.1.15 BITE. BITE shall be implemented within the equipment which shall permit the isolation of a fault to a single module or assembly. All BITE controls and indicators shall be mounted on the front panel.

3.1.16 Illumination of operating controls. Illumination of equipment controls is not required.

3.1.17 Equipment operating life. The equipment shall have a total operating life of at least 50,000 hours.

3.1.18 Time meters. One time meter shall be provided on the front panel of the equipment which shall indicate the total elapsed time that prime power is applied to the equipment. The time meter shall be in accordance with MIL-M-7793.

3.1.19 Service conditions. The equipment shall conform to the operating requirements of this specification under any combination of environmental conditions specified herein.

3.1.20 Nuclear effects. Nuclear effects shall be as specified in 3.1.20.1 and 3.1.20.2.

3.1.20.1 Electromagnetic pulse (EMP) hardening. The EMP threat levels for the equipment shall be as specified in the nuclear environment statement, PME 116, 11019/SAL, serial 00684 of 5 March 1976. The equipment shall survive the specified EMP threat level environment without degradation of any performance characteristics of the equipment, as specified herein, or require any repair actions necessary to restore full performance capabilities of the equipment. Uninterrupted operation of the equipment during the EMP is not required. The equipment shall be designed and constructed in accordance with the recommendations specified in the Index of recommended practices paragaph of DNA 2114 H-1. The engineering practices in the Some EMP considerations of engineering practices chapter, the Shielding, grounding and cabling practices chapter, and the Terminal protection devices and their application chapter of DNA 2114 H-1 pertinent to the hardening of equipment to EMP shall be utilized in the design of the modem.

3.1.20.2 Transient radiation effects hardening. The equipment shall survive the transient radiation effects on electronics (TREE) exposure levels specified in the Nuclear Environment statement, PME 116, 11019/SAL, serial 00684 of 5 March 1976. The equipment is not required to operate while subjected to the TREE environment. The TREE hardening techniques used in the design and fabrication of the equipment shall comply with DNA 1420 H, TREE Handbook. Exposure of the equipment, without the benefit of shielding from nuclear radiation other than that employed in the design and fabrication of the equipment, shall not result in causing either direct permanent damage effects or in requiring any repair actions necessary to restore full performance capabilities of the equipment as specified herein.

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3.2 Detailed requirements. Detailed requirements shall be as specified in 3.2.1 through 3.2.1.18.

3.2.1 Input power. The equipment shall operate from a nominal alternating current (AC) power source of 115 volts (V) to 230 V, 60 Hz, single-phase Type I, in accordance with DoD-STD-1399, Section 300. The input power required to operate equipment in any mode shall be less than 1.4 kilovoltamperes at 0.8 power factor.

3.2.2 Transmit and receive turn-around time. The transmit and receive turn-around time for the equipment shall not exceed 7 milliseconds (ms) when the equipment is being operated in the LOS mode.

3.2.3 Emissions. AN/WSC-3(V) versions shall provide transmission and reception of 6A3, 60A3, 16F3, 2.5F1, 0.15F9, 0.6F9, 2.4F9, 4.8F9, 9.6F9, 11F9, 19.2F9, 25F9, and 60F9 types of emissions, in accordance with TABLE II. Bandwidths of the F1 type of emissions, as specified in APPENDIX A, Designation of emissions and bandwidth, of MIL-STD-188, shall be determined in accordance with the requirements of this specification for FSK data modems.

3.2.4 Frequency coverage. The equipment shall be capable of operation, both transmit and receive as specified herein, on 7000 discrete radio frequency channels. The radio frequency channels shall be spaced at integral 25-kilohertz (kHz) increments over the frequency range of 225.000 MHz to 399.975 MHz.

3.2.4.1 Receive frequency off-set. Both the SATCOM and LOS versions of the radio set shall be delivered with a switch to quick change to the frequency off-sets specified in TABLE IV.

TABLE IV. Frequency off-set.

Satellite	Receive frequency off-set (below transmit frequency) (in MHz)
Gapfiller	53.6
TACSAT/LES-9	53.8
FLTSAT 500 kHz	33.6
FLTSAT 25 kHz	41.0

A label shall be mounted near the frequency off-set select switch identifying the proper switch positions for the frequency offsets specified in TABLE IV.

3.2.4.1.1 Receive frequency offset expansion. The expansion of the receive frequency offset capability shall allow the selection of at least six different receive frequency offsets from a control internally mounted in the equipment. While transmitting on any integral 25-kHz frequency from 290 MHz to 315 MHz in the SATCOM mode, the equipment shall be capable of receiving on any selected receive frequency off-set. All selectable receive frequency offsets shall be on integral 25-kHz frequencies within a 16-MHz receive band. The center frequency of the 16-MHz receive band shall be capable of being off-set on any integral 25-kHz frequency from 248 MHz to 257 MHz.

3.2.4.2 Frequency calibration resolution. The frequency adjustment control for the internal standard of the equipment shall be periodically calibrated or set within 5 parts in 10^8 of any designated frequency.

3.2.4.3 Frequency stability. The frequency stability of the equipment, with respect to the internal frequency standard, shall be within 1 part in 10^8 over 100 ms, 1 part in 10^7 per day, 2 parts in 10^7 during the first 30 days after calibration. The frequency stability shall not be degraded by more than 1 part in 10^7 for each 30-day period thereafter.

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3.2.4.4 Internal frequency generation. All equipment internal frequencies for transmit and receive shall be synthesized from an internal frequency standard of 1.0 MHz or 5 MHz except that in receiver operation an injection signal for the second mixer shall be from an oven or temperature stabilized voltage control crystal oscillator (VCXO), which shall be locked to the incoming signal in the data decoding modes but which shall operate as a fixed-frequency oscillator in the AM and FM receive modes. A buffered output from the internal frequency standard, protected from short circuit and extraneous voltages, shall be provided to a Type BNC connector mounted on the front panel.

3.2.4.5 External frequency injection. The equipment shall permit the selection of either the internal frequency standard or an external frequency standard. The internal standard shall be capable of being bypassed to permit use of an external standard. The external reference source shall be supplied at 0.1 volts root-mean-square (Vrms) to 3.0 Vrms level into 50 ohms. A Type BNC connector shall be mounted on the rear of the equipment for the injection of the external frequency source. When the equipment is operating with an external frequency source, the equipment shall exhibit the frequency stability of the external source.

3.2.4.6 Frequency accuracy. The frequency accuracy required in the AM and FM modes of the transmitter or receiver shall be within at least one part in 10^6 of the selected frequency.

3.2.5 Channel selection system. All channels of the transmitter and receiver shall be available without the necessity of changing any components and manually adjusting any circuits. When a channel has been selected, both the transmitter and receiver circuits shall adjust automatically for operation on the same frequency selected, or as specified herein for SATCOM. The time elapsing between the instant the channel has been selected and the instant the equipment is in an operative condition on the selected channel shall not exceed 5 ms.

3.2.5.1 Spontaneous channeling. When the equipment is operating, there shall be no spontaneous change of channels or channeling when the equipment is subjected to any combination of service conditions in accordance with MIL-E-16400 as specified herein.

3.2.5.2 Preset channel memory. A preset channel memory device for storing 20 of the 7000 available channels shall be provided as part of the equipment for all radio versions except the V(8) configuration. The preset device shall permit the selection of any of the preset channels by means of a single selector marked from 1 to 20 inclusive. A maximum of 15 seconds shall be required for presetting a channel . The presetting device and functions shall be so designed that the operators can readily determine preset channel and manual operational modes.

3.2.5.2.1 Channel frequency chart. A preset channel frequency chart shall be provided on the front control panel of the equipment and remote control unit for recording in pencil the frequency corresponding to each preset channel number. The surface of the chart shall be such that pencil entries may be clearly recorded and erased in the order of 1000 times.

3.2.6 Transmitter performance. Unless otherwise specified herein, the transmitter shall comply with the specified performance requirements, when tested under service test conditions, within 10 seconds after the equipment is turned from STANDBY to ON. Where a specific degradation limit has not been stated, the degradation shall not exceed 1 dB for requirements specified in dB and 5 percent for all other parameters. The equipment shall be capable of continuous transmit duty cycle. A standby switch will be provided to permit operation of ovens while the remaining portion of the equipment is deactivated. A thermal switch shall be provided to activate the blower in standby condition if the temperature increase warrants.

3.2.6.1 Radio frequency (RF) power output. The transmitter unmodulated RF power output shall be a minimum of 100 W (+50 decibels referred to one milliwatt (dBm)) for FM and 30 W (+44 dBm) for AM into a 50-ohm resistive load voltage standing wave ratio (VSWR) of 1:1. The transmitter output power amplifiers shall have a minimum overall efficiency of 25 percent.

3.2.6.1.1 RF power output control. The transmitter RF power output shall be continuously adjustable in all modes except AM from full rated output to 1 W output by a power level control mounted on the front panel.

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3.2.6.1.2 RF power output indicator. A reflectometer shall be included in the equipment to provide incident and reflected power readings on a front panel meter. The reflectometer meter shall be separate from the BITE meter. The accuracy of the reflectometer meter readings at full rated power (30 W and 100 W) shall be within ±1 dB of the actual RF power output from the equipment. The full scale meter reading shall be 23 decibels referred to one watt (dBW) and 16 dBW.

3.2.6.1.3 RF power output keying. The transmitter RF power output to the antenna group shall be delayed 40 ms +10 ms after keying when the equipment is operating in the SATCOM 25 kHz, or any of the SATCOM OFFSET kHz communication modes. The equipment shall not be damaged in any way if the transmitter is keyed while the equipment is channeling.

3.2.6.2 Mismatched load. The equipment shall be protected from damage due to shorts or opens in the interconnecting RF transmission line from the radio set to the antenna or due to a VSWR greater than 4 to 1. When the transmitter is connected to any impedance producing a VSWR of 2.5 to 1, the unmodulatd RF power output shall not be degraded more than 1 dB for FM or AM. Between 2.5 to 1 and 4 to 1, the equipment may operate at reduced power and be protected from damage.

3.2.6.3 Spurious radiation. The spurious radiation from the equipment referenced to the rated RF power output of the transmitter shall not exceed -50 dB at frequencies up to ±5 MHz from the carrier frequency. From ±5 MHz from the carrier frequency to ±5 percent of the carrier frequency, the spurious level limits shall vary linearly from -50 dB to -80 dB. Beyond ±5 percent of the carrier frequency, the spurious level limit shall be -80 dB.

3.2.6.4 Harmonic attenuation. All harmonics and subharmonics shall be attenuated at least 60 dB below the full power output.

3.2.6.5 AM harmonic distortion. With any audio input frequency between 300 Hz and 3500 Hz, and with the audio input adjusted for a level equal to 90 percent modulation at 1 kHz, the total harmonic distortion generated by the equipment, including the modulator, shall not exceed 5 percent while the equipment is being operated in the transmit mode at full rated RF power output. Voice clipping shall be disabled during test for this requirement, by means of a clipper disable switch within the transceiver.

3.2.6.6 Intermodulation distortion for each tone. With any two equal level audio input frequencies between 300 Hz and 3500 Hz and with each audio input adjusted for 45 percent modulation, the intermodulation distortion products at rated power output shall not exceed -40 dB with reference to a detected output voltage obtained from the carrier 90 percent modulated with a 1000-Hz tone. Voice clipping shall be disabled during test for this requirement, by a clipper disable switch within the transceiver.

3.2.6.7 FM noise. The total peak deviation developed by internal noise of the equipment, including the modulator, shall not exceed 75 Hz peak while the equipment is operating in the transmit mode at full rated RF power output with no modulation applied. The FM noise measurement shall be made in a 3500-Hz bandwidth.

3.2.6.8 Secure voice interface characteristics. Secure voice interface characteristics shall be as specified in 3.2.6.8.1 through 3.2.6.8.3.

3.2.6.8.1 AM diphase mode. The AM diphase mode shall be as specifed in 3.2.6.8.1.1 through 3.2.6.8.1.2.3.

3.2.6.8.1.1 V(3) and V(7) configurations. See 1.3, 1.4, and 1.5 for configuration explanations. The equipment shall provide the SAS and TSEC/KY-58 and TSEC/KY-8 interface characteristics of 3.2.6.8.1.1.1 through 3.2.6.8.1.1.3.

3.2.6.8.1.1.1 Frequency response. The frequency response shall be in accordance with the applicable CSEEB document requirements specified in 3.2.8.5.4.

3.2.6.8.1.1.2 Input impedance. The input impedance shall be 600 ohms ±15 percent balanced with respect to ground.

3.2.6.8.1.1.3 Input level. An internal adjustment shall be provided to set the AM of the carrier to 95 percent with any input signal level between -8 dBm and +8 dBm.

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3.2.6.8.1.2 V(2) and V(6) configurations. The equipment shall provide the standard TSEC/KY-58 and TSEC/KY-8 interface characteristics of 3.2.6.8.1.2.1 through 3.2.6.8.1.2.3.

3.2.6.8.1.2.1 Frequency response. The frequency response shall be in accordance with the applicable CSEEB document requirements specified in 3.2.8.5.4.

3.2.6.8.1.2.2 Input impedance. The input impedance shall be 2000 ohms nominal unbalanced with respect to ground.

3.2.6.8.1.2.3 Input level. An internal adjustment shall be provided to set the amplitude modulation of the carrier to 95 percent with any sinusoidal input signal level between 6 V and 14 V peak-to-peak (P-P).

3.2.6.8.2 FM baseband mode. The FM baseband mode shall be as specified in 3.2.6.8.2.1 through 3.2.6.8.2.1.3.

3.2.6.8.2.1 V(3) and V(7) configurations. The equipment shall provide the SAS, TSEC/KY-58 and TSEC/KY-8 interface characteristics of 3.2.6.8.2.1.1 through 3.2.6.8.2.1.3.

3.2.6.8.2.1.1 Frequency response. The audio power, at a frequency between 10 Hz and 8 kHz, resulting from demodulation of the frequency modulated transmitter output, shall be within +0 dB and -3 dB of the maximum demodulated audio power measured using a peak deviation of 5.5 kHz.

3.2.6.8.2.1.2 Input impedance. The input impedance shall be 600 ohms ±15 percent balanced with respect to ground.

3.2.6.8.2.1.3 Input level. The equipment shall provide an internal adjustment to set the peak deviation of the carrier frequency from 2.5 kHz to 7.5 kHz with a 1-kHz input signal at any input level between -8 dBm and +8 dBm.

3.2.6.8.2.2 V(2) and V(6) configurations. The equipment shall provide the standard TSEC/KY-58 and TSEC/KY-8 interface characteristics of 3.2.6.8.2.2.1 through 3.2.6.8.2.2.3.

3.2.6.8.2.2.1 Frequency response. The audio power, at a frequency between 10 Hz and 8 kHz, resulting from demodulation of the frequency modulated transmitter output, shall be within +0 dB and -3 dB of the maximum demodulated audio power measured using a peak deviation of 5.5 kHz.

3.2.6.8.2.2.2 Input impedance. The input impedance shall be 2000 ohms nominal unbalanced with respect to ground.

3.2.6.8.2.2.3 Input level. The equipment shall provide an internal adjustment to set the peak deviation of the carrier from 2.5 kHz to 7.5 kHz with a 1-kHz input signal at any level between 6 and 14VP-P.

3.2.6.8.2.3 Radiated power bandwidth. A minimum of 99 percent of the energy of the frequency modulated transmitter signal, having a deviation ratio of 0.7 and modulated by a pseudo-random sequence at 16 kilobits per second (kbps) shall be contained within a 25-kHz bandwidth.

3.2.6.8.2.4 Deviation ratio. The nominal deviation ratio is the ratio of the Peak-to-Peak (ΔF) deviation of the transmit signal to a 16 kbps FM rate and shall be 0.7 (deviation ratio = $\Delta F/FM$).

3.2.6.8.3 Keyline. A ground, applied to the equipment keyline, shall cause the transmitter to Key-ON; an open circuit shall cause the transmitter to Key-OFF.

3.2.6.9 Plain text interface characteristics. Plain text interface characteristics shall be as specified in 3.2.6.9.1 and 3.2.6.9.2.

3.2.6.9.1 AM plain text mode. The equipment, V(1), V(2), V(3), V(5), V(6), and V(7) configurations, shall provide AM plain text interface characteristics in accordance with the applicable requirements of 3.2.6.5, 3.2.6.6, 3.2.8.5.1, 3.2.8.5.2, and 3.2.8.5.3.

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3.2.6.9.2 FM plain text mode. The equipment, V(1), V(2), V(3), V(6), and V(7) configurations, shall provide FM plain text interface characteristics in accordance with the applicable requirements of 3.2.8.6.1, 3.2.8.6.2, 3.2.8.6.4 and 3.2.8.6.5.

3.2.6.10 LINK 11 (transmit). LINK 11 (transmit) shall be as specified in 3.2.6.10.1 through 3.2.6.10.1.4.

3.2.6.10.1 V(2), V(3), V(6), and V(7) configurations. The equipment shall provide the LINK 11 interface characteristics of 3.2.6.10.1.1 through 3.2.6.10.1.4.

3.2.6.10.1.1 Frequency response. The audio power, at a frequency between 300 Hz and 3500 Hz, resulting from demodulation of the frequency modulated transmitter signal, shall be within +0 dB and -3 dB of the maximum demodulated audio power measured using a peak deviation of 8 kHz.

3.2.6.10.1.2 Input impedance. The input impedance shall be 600 ohms ±15 percent balanced with respect to ground.

3.2.6.10.1.3 Input level. The equipment shall provide an internal adjustment to set the peak deviation of the carrier frequency to 20 kHz with any sinusoidal input signal level between -3 dBm and +15 dBm.

3.2.6.10.1.4 LINK 11 key. The transmitter shall key ON with 6 V ±1 volt direct current (VDC) at less than 3 mA applied as a phantom signal to the balanced audio input terminals. The transmitter shall key OFF when 0.75 VDC or less is applied.

3.2.7 Receiver performance. Unless otherwise specified herein, the equipment shall conform to the specified performance requirements, when tested under service conditions within 10 seconds after the set is turned from STANDBY to ON. Where a specific degradation limit has not been stated, the degradation shall not exceed 1 dB for performance requirements specified in dB, and 5 percent for all measurements. The equipment shall be capable of continuous receive duty cycle.

3.2.7.1 Selectivity. The response of the receiver to signals less than ±15 kHz from the selected receiver frequency shall not be attenuated more than 3 dB referenced to the response at the center frequency. The response of the receiver to signals more than ±50 kHz from the selected receiver frequency shall be attenuated more than 60 dB referenced to the response at the center frequency. The gain variation over the passband shall not vary more than ±1.5 dB referenced to the amplitude response at the selected frequency. For operation with the TSEC/KY-8, the intermediate frequency (IF) bandwidth can be widened if necessary.

3.2.7.2 Sensitivity. The sensitivity of the receiver shall be such that when a signal generator having a source impedance of 50 ohms and an open circuit RF output of not more than 3.0 microvolts (μV) (-97.5 dBm) modulated ±2500 Hz at 1000 Hz for FM or 3.5 μV (-96.0 dBm) with 30 percent modulation at 1000 Hz AM is connected to the receiver antenna terminals through a 50-ohm transmission line, the receiver shall produce an audio output with a 10 dB signal-plus-noise to noise (S+N/N) ratio or greater.

3.2.7.3 Noise figure. The noise figure of the receiver shall not exceed 13 dB over the operating frequency range of the receiver.

3.2.7.4 Overload. Performance of the receiver, as specified herein, shall be obtained over the input signal level variations of 3.5 μV to 100 millivolts (mV) rms. There shall be no damage or degradation to the specified performance subsequent to the application of any RF signal, up to 100 Vrms to the receiver antenna input terminals when the receiver is either in an operating or nonoperating condition.

3.2.7.5 Spurious response. The receiver response to spurious signals displaced from the selected receiver center frequency (f_0) by less than ±20 MHz shall be attenuated not less than 80 dB with respect to the received signal. The receiver response to spurious signals less than 1.4 gigahertz (GHz) and displaced from the selected f_0 by ±20 MHz or more shall be attenuated at least 100 dB with respect to the desired signal. The receiver response to spurious signals greater than 1.5 GHz shall be attenuated 90 dB relative to the response to the desired frequency. At frequencies of one-half of the receiver operating frequencies, the spurious responses shall be attenuated at least 80 dB.

3.2.7.6 Image rejection. The response of the receiver to image frequencies shall be attenuated at least 70 dB with respect to the received signal from 225 MHz to 330 MHz and at least 65 dB from 330 MHz to 400 MHz.

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3.2.7.7 IF rejection. The response of the receiver to signals at the IF amplifier frequencies shall be attenuated at least 100 dB with respect to the selected f_o.

3.2.7.8 AM harmonic distortion. With any audio frequency between 300 Hz and 3500 Hz modulating a -60 dBm carrier signal applied to the receiver input, the total harmonic distortion generated by the equipment, including the demodulator, shall not exceed 5 percent with reference to a detected output voltage from -60 dBm carrier signal 90 percent modulated while the equipment is being operated in the receive mode.

3.2.7.9 Intermodulation distortion. With a carrier input, modulated 90 percent with any two equal level input tones between 300 Hz and 3500 Hz, the intermodulation products shall not exceed -40 dB with reference to a detected output voltage obtained from the carrier 90 percent modulated with a 1000 Hz tone.

3.2.7.10 FM noise. The total peak deviation developed by internal noise of the equipment, including the demodulator, shall not exceed 75 Hz while the equipment is operating in the receive mode with a -60 dBm unmodulated carrier applied to the input of the equipment. The FM noise measurement shall be made in a 3500-Hz bandwidth.

3.2.7.11 Automatic gain control (AGC) AM. The receiver shall employ AGC circuitry such that the audio output from the equipment shall not vary more than ±3 dB, referenced to the audio output obtained from a 1-mV antenna input signal modulated 90 percent with a 1000 Hz tone, as the received signal varies from 3.5 µV to 0.1 V.

3.2.7.12 Pulse blanking. Prior to demodulation, inband pulse interference shall be removed by blanking the receiver during the period of the pulse. Techniques which will remove significant energy from the desired receive signal shall not be used. Blanking shall occur without receipt of information from sources external to the equipment. The typical sources of inband pulse interference will be from AN/SPS-43 (series), AN/SPS-37, and AN/SPS-40 radar equipment. Provision shall be made in the equipment to enable or inhibit such blanking action.

3.2.7.13 Squelch and signal presence V(1), V(2), V(3), V(6), and V(7) configurations. Squelch and signal pressure for V(1), V(2), V(3), V(6), and V(7) configurations shall be as specified in 3.2.7.13.1 through 3.2.7.13.3.

3.2.7.13.1 Squelch. The plain and secure text output shall be capable of being squelched. A switch on the front panel of the equipment shall be provided to disable the squelch circuitry.

3.2.7.13.2 Signal presence. The equipment shall provide a contact closure to ground to indicate a signal, within the receiver passband, that is greater than 2 dB above the plain and secure squelch threshold setting. The signal presence contact closure feature shall be functional regardless of the ON-OFF status of the squelch disable switch.

3.2.7.13.3 Response time. The time delay, from the application of a receiver input signal (signal level greater than 6 dB above squelch threshold) to the grounding of the signal presence line, and plain and secure text output, shall be less than 35 ms.

3.2.7.14 Secure voice interface characteristics. Secure voice interface characteristics shall be as specified in 3.5.7.14.1.1 through 3.5.7.14.3.

3.2.7.14.1 AM diphase mode. The AM diphase mode shall be as specified in 3.2.7.14.1 through 3.2.7.14.1.2.3.

3.2.7.14.1.1 V(3) and V(7) configurations. The equipment shall provide the SAS, TSEC/KY-58, and TSEC/KY-8 interface characteristics of 3.2.7.14.1.1.1 through 3.2.7.14.1.1.3.

3.2.7.14.1.1.1 Frequency response. The frequency response shall be in accordance with the applicable CSEEB document requirements specified in 3.2.8.5.4.

3.2.7.14.1.1.2 Output power. The output power shall be adjustable from -8 dBm to +8 dBm when a -60 dBm carrier, 90 percent amplitude modulated by a 19-kHz sinusoidal signal is applied to the antenna input.

3.2.7.14.1.1.3 Output impedance. The output impedance shall be 600 ohms ±15 percent balanced with respect to ground.

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3.2.7.14.1.2 V(2) and V(6) configurations. The equipment shall provide the standard TSEC/KY-58 and TSEC/KY-8 interface characteristics of 3.2.7.14.1.2.1 through 3.2.7.14.1.2.3.

3.2.7.14.1.2.1 Frequency response (R_L = 20 kilohms (kohms) nominal). The frequency response shall be in accordance with the applicable CSEEB document requirements specified in 3.2.8.5.4.

3.2.7.14.1.2.2 Output level (R_L = 20 kohms). The output shall be adjustable from 0.5 VP-P to 2 VP-P minimum range when a -60 dBm signal, 90 percent amplitude modulated by a 19 kHz sinusoidal signal is applied to the antenna input.

3.2.7.14.1.2.3 Output load impedance. The output impedance shall be 20 kohms minimum unbalanced with respect to ground. The output source impedance shall be less than 2 kohms.

3.2.7.14.2 FM baseband mode. The FM baseband mode shall be as specified in 3.2.7.14.2.1 through 3.2.7.14.2.2.3.

3.2.7.14.2.1 V(3) and V(7) configuration. The equipment shall provide the SAS, TSEC/KY-58, and TSEC/KY-8 interface characteristics of 3.2.7.14.2.1.1 through 3.2.17.14.2.1.3.

3.2.7.14.2.1.1 Frequency response. An antenna input signal of -60 dBm, deviated to 5.5 kHz (peak) by a sinusoidal signal between 10 Hz and 10 kHz, shall produce an audio output power that is within +0, -3 dB of the highest audio power with the band.

3.2.7.14.2.1.2 Output power. The audio output power shall be adjustable from -8 dBm to +8 dBm when a -60 dBm carrier, deviated to 5.5 kHz (peak) by an 8-kHz sinusoidal signal, is applied to the antenna input.

3.2.7.14.2.1.3 Output impedance. The output impedance shall be 600 ohms ±15 percent balanced with respect to ground.

3.2.7.14.2.2 V(2) and V(6) configurations. The equipment shall provide the standard TSEC/KY-58 and TSEC/KY-8 interface characteristics of 3.2.7.14.2.2.1 through 3.2.7.14.2.2.3.

3.2.7.14.2.2.1 Frequency response (R_L = 20 kohms nominal). A -60 dBm antenna input signal, deviated to 5.5 kHz (peak) by any sinusoidal signal between 10 Hz and 10 kHz, shall produce an audio output power that is within +0, -3 dB of the highest audio power within the band.

3.2.7.14.2.2.2 Frequency response (R_L = 20 kohms). The audio output power shall be adjustable from 0.5 VP-P to 2 VP-P minimum range when a -60 dBm carrier, deviated to 5.5 kHz (peak) by an 8-kHz sinusoidal signal, is applied to the antenna input.

3.2.7.14.2.2.3 Output load impedance. The output load impedance shall be 20 kohms minimum unbalanced with respect to ground. The output source impedance shall be less than 2 kohms.

3.2.7.14.3 FM sensitivity. A 6 µV (open circuit) antenna terminal input signal, deviated to 5.5 kHz peak by an 8 kHz sinusoidal signal, shall produce at the audio output a (S+N)/N ratio of equal to or greater than 10 dB.

3.2.7.15 Plain text interface characteristics. Plain text interface characteristics shall be as specified in 3.2.7.15.1 through 3.2.7.15.1.1.3.

3.2.7.15.1 AM mode. The AM mode shall be as specified in 3.2.7.15.1.1 through 3.2.7.15.1.1.3.

3.2.7.15.1.1 V(1), V(2), V(3), V(5), V(6), and V(7) configurations. The equipment shall provide the SAS, TSEC/KY-58, and TSEC/KY-8 interface characteristics of 3.2.7.15.1.1.1 through 3.2.7.15.1.1.3.

3.2.7.15.1.1.1 Frequency response. An antenna input signal of -60 dBm amplitude modulated to 30 percent by any sinusoidal signal between 325 Hz and 1000 Hz, shall produce an output power of no more than 1 dB above nor more than 2 dB below a reference audio output power level at 1 kHz; from 1000 Hz to 3450 Hz, the output power shall be within ±1.5 dB of the 1-kHz reference power level.

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3.2.7.15.1.1.2 Output power. The output power shall be adjustable from -8 dBm to +8 dBm when a -60 dBm carrier, amplitude modulated to 90 percent by a 1-kHz sinusoidal signal, is applied to the antenna input.

3.2.7.15.1.1.3 Ouput impedance. The output impedance shall be 600 ohms ±15 percent balanced with respect to ground.

3.2.7.15.2 FM mode. The FM mode shall be as specified in 3.2.7.15.2.1 through 3.2.7.15.2.1.3.

3.2.7.15.2.1 V(2), V(3), V(6), and V(7) configurations. The equipment shall provide the TSEC/KY-58 and TSEC/KY-8 interface characteristics of 3.2.7.15.2.1.1 through 3.2.7.15.2.1.3.

3.2.7.15.2.1.1 Frequency response. The application of a -60 dBm carrier deviated to 5 kHz (peak) by any sinusoidal signal between 325 Hz and 1000 Hz shall produce an output power of no greater than 1 dB above nor more than 2 dB below a reference audio output power level at 1 kHz; from 1000 Hz to 3450 Hz, the output power shall be within ±1.5 dB of the 1-kHz reference power level.

3.2.7.15.2.1.2 Output power. The received audio output power shall be adjustable from -8 dBm to +8 dBm when a -60 dBm carrier, deviated to 5 kHz (peak) by a 1-kHz sinusoidal signal, is applied to the antenna input.

3.2.7.15.2.1.3 Output impedance. The output impedance shall be 600 ohms ±15 percent balanced with respect to ground.

3.2.7.16 LINK 11 (receiver). LINK 11 (receiver) shall be as specified in 3.2.7.16.1 through 3.2.7.16.3.

3.2.7.16.1 V(2), V(3), V(6), and V(7) configurations. The equipment shall provide the LINK 11 interface characteristics of 3.2.7.16.1.1 through 3.2.7.16.1.3.

3.2.7.16.1.1 Frequency response. An antenna input signal of -60 dBm, deviated to 8 kHz (peak) by any sinusoidal signal between 300 Hz and 3500 Hz shall produce an audio output power that is within +0, -3 dB of the highest audio power within the band.

3.2.7.16.1.2 Output power. The audio output power shall be adjustable from 3 dBm to +15 dBm when a -60 dBm carrier, deviated to 20 kHz (peak) by a 1-kHz sinusoidal signal, as applied to the antenna input.

3.2.7.16.1.3 Output load impedance. The output load impedance shall be 600 ohms ±15 percent balanced with respect to ground.

3.2.8 Modem. The modem shall be as specified in 3.2.8.1 through 3.2.8.9.4.

3.2.8.1 General description. The equipment shall provide a modem to convert voice, 75-baud teletype, and 300-baud, 1200-baud, 2400-baud, 4800-baud, and 9600-baud digital data into a modulated 70-MHz carrier as specified herein. The modem shall also demodulate a 70-MHz IF signal and format the information into voice, 75-baud teletype, and 300-baud, 1200-baud, 2400-baud, 4800-baud, and 9600-baud digital data.

3.2.8.2 General requirements. The modem shall be capable of synchronizing and automatically fine tuning to the frequency of the received signal as well as track drift, offset, and rapid changes in the received signal due to Doppler effects. The modem shall be designed to comply with Section 4 of MIL-STD-188, except as specified herein, for the types of emissions specified in 3.2.3. All modem performance measurements shall be made with the modem in a back-to-back configuration, except as otherwise specified herein. AM may be accomplished at final frequency if desired, in which case the AM requirements specified in 3.2.8.5 through 3.2.8.5.4.2 apply to the overall transmitter.

3.2.8.3 Handset (voice operation). A connector shall be provided on the front panel for voice operation of the equipment with an H-169/U handset. The equipment shall comply with the operation and interface requirements of the H-169/U handset in accordance with drawing RE-F-2688036. The contractor shall not supply an H-169/U as part of the equipment.

3.2.8.4 Audio output. The audio output shall be as specified for tactical UHF high performance equipment in MIL-STD-188, Section 4, unless otherwise specified herein.

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3.2.8.5 AM. AM shall be as specified in 3.2.8.5.1 through 3.2.8.5.4.2.

3.2.8.5.1 Carrier noise level. The detected output voltage obtained from an unmodulated carrier shall be at least 40 dB below the detected output voltage obtained from the carrier 90 percent modulated within a 1000-Hz tone.

3.2.8.5.2 Modulation. When a zero dBm audio signal at any frequency between 300 Hz and 3500 Hz is applied to the audio input circuit, the RF carrier output, as rated output level, shall be capable of being modulated at least 95 percent.

3.2.8.5.3 Attenuation versus frequency narrowband mode (audio). The nominal 3 dB points for the transmit radio channel shall be at 300 Hz and 3500 Hz. For all frequencies between 325 Hz and 1000 Hz, the attenuation shall be within the limits of -1 dB and +2 dB, with respect to the attenuation at 1000 Hz. For all frequencies between 1000 Hz and 3450 Hz, the attenuation shall be within the limits of ±1.5 dB, with respect to the attenuation at 1000 Hz.

3.2.8.5.4 Wideband mode (diphase). The wideband mode (diphase) shall be as specified in 3.2.8.5.4.1 and 3.2.8.5.4.2.

3.2.8.5.4.1 TSEC/KY-8. The requirement for an auxiliary wideband mode, diphase for 225 MHz to 400 MHz UHF equipment as specified in MIL-STD-188, Section 4, is applicable to this equipment. The equipment shall be compatible with the TSEC/KY-8 in the wideband mode of operation in accordance with CSEEB-6.

3.2.8.5.4.2 TSEC/KY-58. The equipment shall provide secure and plain text interfaces, compatible with TSEC/KY-58 equipment, in accordance with CSEEB-32 and CSESD-14.

3.2.8.6 FM. FM shall be as specified in 3.2.8.6 through 3.2.8.6.5.

3.2.8.6.1 FM voice modem. The equipment shall be capable of being frequency modulated by a voice or multitone signal. A peak-to-peak deviation of 16 kHz shall be considered 100 percent modulation when modulated with a 1000-Hz tone throughout the range specified in 3.2.8.6.4. Pre-emphasis and de-emphasis shall not be utilized.

3.2.8.6.2 FM modem capability. The FM voice capability shall be as specified for a 4 kHz-channel in accordance with MIL-STD-188, Section 4, except as otherwise specified herein.

3.2.8.6.3 Voice frequency ringing. The use of voice frequency ringing or supervisory signals shall not be required as part of this equipment.

3.2.8.6.4 Audio input level. The modulator shall be adjustable so that any audio frequency level from -4 dBm to +16 dBm shall be capable of producing 100 percent modulation.

3.2.8.6.5 Linearity. The FM linearity shall be such that with both a 1600-Hz tone adjusted to 4-kHz deviation and a 600-Hz tone adjusted to 4-kHz deviation applied to the FM input, the output through the range of 300 Hz to 3500 Hz (600 Hz and 1600 Hz excluded) shall be at least 34 dB below a single test tone deviating the carrier 8 kHz.

3.2.8.7 Digital data. Digital data shall be as specified in 3.2.8.7.1 through 3.2.8.7.3.4.

3.2.8.7.1 Digital data operation. The digital data portion of the modem shall operate at 75 baud FSK and 75 baud, 300 baud, 1200 baud, 2400 baud, 4800 baud, and 9600 baud PSK in accordance with MIL-STD-188, Section 7, except as specified herein. The modem shall be compatible for both high level and low level standard digital interface operation specified in MIL-STD-188, Section 7. The equipment shall be compatible for operation with the C-1004/SG control unit in accordance with NAVSEA 0967-LP-143-4010, while operating in the digital data mode.

3.2.8.7.1.1 Digital data clock. Transmit and receive bit timing shall be provided separately at discrete outputs and combined at a single output. The combined output shall be used for operation with half duplex key generators such as the TSEC/KG-36.

3.2.8.7.2 FSK data modem. The FSK portion of the modem shall be capable of 75-baud asynchronous nonisochronous teletype operation. The bandwidth of the emission shall be optimized based on overall equipment requirements including both satellite and LOS operation.

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3.2.8.7.2.1 Demodulator dynamic range. The demodulator shall operate over a minimum dynamic range of 70 dB without degradation of performance as specified herein.

3.2.8.7.2.2 One-way reversible operation. The one-way reversible operation requirement of MIL-STD-188 is not applicable to this modem.

3.2.8.7.2.3 Frequency acquisition and tracking range. The frequency acquisition and tracking range of the modem for the 75-baud rate shall be ±500 Hz minimum. The tracking loop bandwidth shall be sufficient to track doppler rates of up to 39.2 Hz per second.

3.2.8.7.2.4 Bit error rate (BER) (white gaussian noise). The modem performance-related BER versus E_b/N_o for 75 baud FSK operation shall be:

E_b/N_o (dB)	BER (maximum)
15.0	1 in 10^5
13.0	1 in 10^3

3.2.8.7.3 PSK data modem. The PSK modulator portion of the modem shall convert isochronous serial digital data, in accordance with MIL-STD-188, Section 7, to a biphase modulated 70-MHz carrier. The PSK demodulator portion of the modem shall convert a biphase modulated 70-MHz signal into isochronous serial digital data and synchronous clock signals.

3.2.8.7.3.1 Operating requirements. The modem shall operate in a simplex mode at 75-baud, 300-baud, 1200-baud, 2400-baud, 4800-baud, and 9600-baud rates only.

3.2.8.7.3.2 Demodulator dynamic range. The dynamic range of the demodulator shall be the same as specified in 3.2.8.7.2.1.

3.2.8.7.3.3 Demodulator signal alarm. The provision of a demodulator signal alarm required by MIL-STD-188 is not applicable to this modem.

3.2.8.7.3.4 Signal acquired indicator. A signal acquired circuit for external control shall be provided. The output level shall be ±6 VDC in accordance with MIL-STD-188, Section 7. When the carrier is acquired the output shall be +6 VDC.

3.2.8.8 75-bps FSK data modem. The 75-bps FSK data modem shall be as specified in 3.2.8.8.1 through 3.2.8.8.3.

3.2.8.8.1 Acquisition time. The acquisition time for 75-bps data shall be 8.3 seconds, maximum, for a 90 percent probability at 11 dB E_b/N_o differential phase shift key (DPSK).

3.2.8.8.2 Frequency acquisition and tracking range. The frequency acquisition and tracking range of the modem for the 75-bps rate shall be ±1 kHz minimum. The tracking loop bandwidth shall be sufficient to track Doppler rates of up to 39.2 Hz per second.

3.2.8.8.3 BER (white gaussian noise). The modem performance related BER versus E_b/N_o for 75-bps operation shall be:

E_b/N_o (PSK)	E_b/N_o (DPSK)	BER (maximum)
13 dB	14 dB	1 in 10^5
10 dB	11 dB	1 in 10^3

3.2.8.9 300-bps, 1200-bps, 2400-bps, 4800-bps, and 9600-bps PSK modem. The 300-bps, 1200-bps, 2400-bps, 4800-bps, and 9600-bps PSK modem shall be as specified in 3.2.8.9.1. through 3.2.8.9.4.

3.2.8.9.1 Frequency acquisition. The demodulator shall conform to the BER requirements of 3.2.8.8.3 and 3.2.8.9.2 with a frequency uncertainty of ±700 Hz. The tracking loop bandwidth shall be sufficient to track Doppler rates up to 39.2 Hz per second.

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3.2.8.9.2 BER (white gaussian noise). The modem performance related BER versus E_b/N_o for 300-bps, 1200-bps, 2400-bps, 4800-bps, and 9600-bps shall be:

bps	E_b/N_o (PSK)	BER (maximum)	E_b/N_o (DPSK)
300 to 1200 to 2400	12 dB	1 in 10^5	13 dB
4800 to 9600	8.5 dB	1 in 10^3	9.5 dB

3.2.8.9.3 Acquisition time. The acquisition time for 2400 bps data shall be 40 ms, maximum, for a 90 percent probability at 9.5 E_b/N_o DPSK.

3.2.8.9.4 Error rate measurement tolerance. The error rate measurement tolerance shall be 0.5 dB for all data rates.

3.2.9 External modem interface. The equipment shall provide an external modem interface at 70 MHz as specified in a and b:

a. Input to set from external modem:
   1. Frequency 70 MHz
   2. Bandwidth (3 dB) ±250 kHz
   3. Power level 0 dBm ±1 dB
   4. Impedance 50 ohms
   5. VSWR 1.5:1 (maximum)
   6. Type RF connector BNC

b. Output from set to external modem:
   1. Frequency 70 MHz
   2. Bandwidth (3 dB) ±250 kHz
   3. Load impedance 50 ohms
   4. Load VSWR 1.5:1 (maximum)
   5. Type RF connector BNC

3.2.10 Antenna group interface requirements. The equipment shall be compatible for operation with antennas which have the interface requirements specified in a through d:

a. The RF connector for the antenna of the equipment shall be Type N and it shall be located on the rear of the cabinet.

b. The equipment shall provide a voltage of +12 VDC at 200 mA for operation of external preamplifier bypass relays when the AN/WSC-3 is in transmit mode.

c. The equipment shall provide a set of SPOT dry contacts, isolated from ground, for operation of external antenna selection relays. Current rating shall be 300 mA, minimum. One contact shall be closed in the SATCOM mode; the other contact shall be open in the SATCOM mode.

d. The equipment shall provide a keyline interrupt line to permit use of an external interlock relay which will prevent transmitter keying when antenna circuits are disconnected.

3.2.11 Local and remote operation. Local and remote operation shall be as specified in 3.2.11.1 through 3.2.11.4.

3.2.11.1 Local operation. All operational functions of the equipment shall be controlled from the front panel of the set when the LOCAL/REMOTE selector is in the local position. It shall not be possible to change or override any operational or control function of the equipment from any remote control station while the LOCAL/REMOTE selector is in the LOCAL position.

3.2.11.2 Remote station operation. The equipment shall be compatible for push-to-talk (PTT) operation from a C-1138( )/UR remote control station and shall comply with the interface requirements of MIL-C-22954, except the remote START and continuous wave (CW) key functions are not required. The equipment shall provide 115 V single-phase, 60 Hz power to the C-1138( )/UR for the equipment ON lamp indicator. The PTT keyline shall be a ground in transmit. The external voltage supplied to the remote station shall be -12 VDC at 7 ampere. A relay contact closure (one side grounded) and a ±12 VDC signal at 80 mA shall also be provided or carrier ON indication.

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3.2.11.3 Control-indicator, C-9351/WSC-3. The C-9351/WSC-3 shall be provided as part of each SATCOM AN/WSC-3(V) equipment ((V)1, (V)2, (V)3 versions). C-9351/WSC-3 units to be provided with LOS versions shall be in the quantity as specified in the contract (see 6.2.1). The LOS equipment shall be capable of full operation without the C-9351/WSC-3. The C-9351/WSC-3 shall be capable of controlling the operation of all versions of the RT-1107 through 457.2 m (1500 feet) of TTRS (1119 cm) cable, except the serial control version which does not utilize the C-9351/WSC-3. The connector for the interconnecting cable shall be mounted on the bottom of the C-9351/WSC-3 and the rear of the RT-1107. The H-169/U handset shall be used with the C-9351/WSC-3. The design of the C-9351/WSC-3 shall permit bulkhead mounting. The following controls and indicators specified in a through f shall be incorporated in the C-9351/WSC-3:

a. Preset channel selection
b. Modulation mode selection
c. Handset volume control (H-169/U handset)
d. EQUIP ON indicator (only when equipment is remoted)
e. CARRIER ON and DATA SIGNAL ACQUIRED indicator
f. MANUAL FREQ SELECTED and SATCOM MODE indicator for AN/WSC-3(V)5, (V)6, and (V)7 equipments, the C-9351/WSC-3 shall have a small area provided on the channel identification plate to permit the installer to identify that the remote control unit is used with LOS radios only.

3.2.11.4 Remote modem status indication. A +28 VDC signal shall be provided to drive remote modem status indicator lamps (that is, CARRIER ON and SIGNAL ACQUIRE). The current rating shall be enough to drive up to three positions with two lamps each.

3.2.12 LINK 11 select. The equipment shall provide a control line to enable the LINK 11 mode. The control line shall be functional only in FM mode. A ground applied to the LINK 11 select line shall enable LINK 11 mode. An open circuit shall enable FM plain and secure capability.

3.2.13 AM (diphase) and FM (baseband) control. The equipment shall provide a transistor-transistor logic (TTL) 0 (0 V nominal) command to the TSEC/KY-58 equipment to indicate that AM mode has been selected. A/TTL 1 (+5V nominal) command shall indicate the selection of FM mode.

3.2.14 Secure and plain text control. A control line shall be provided for the V(3) and V(7) configurations that a) when grounded shall enable transmission and reception of secure text signals and b) when open circuited shall enable the transmission and reception of plain text signals.

3.2.15 BER FM secure text mode. The BER measured using two equipments in a back-to-back configuration shall be less than 1 in 10^3 under the conditions specified in a through c:

a. The transmitter is modulated by a 16-kbps pseudo-random sequence
b. The transmitter nominal deviation ratio (specified in 3.2.6.8.2.4) is 0.7
c. The receiver input signal level is 6 µV (open circuit) minimum

3.2.16 AN/WSC-3(V)8 characteristics. The AN/WSC-3(V)8 shall be identical to the AN/WSC-3(V)7 specified herein, except as specified in 3.2.16.1 through 3.2.16.1.3.

3.2.16.1 AN/WSC-3(V)8 serial controlled system. The serial controlled equipment is shown in FIGURE 1 and FIGURE 2. Data bus lines shall be connected directly to the AN/WSC-3(V) via the formerly used preset channel select lines. In order to control mode on a LOS AN/WSC-3(V), jumper wires shall be installed in the interconnecting cables between connectors J2 and J3 at the back of the case.

3.2.16.1.1 Data interface and message format requirements. The digital data interface shall conform to the Standard interface (low level) paragraph of MIL-STD-188.

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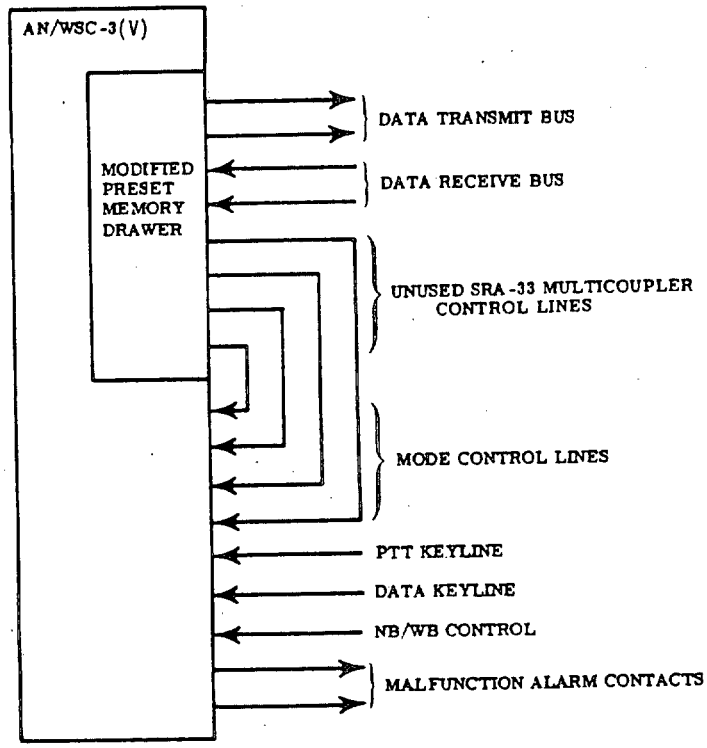


FIGURE 1. AN/WSC-3(V)8 RSC system.

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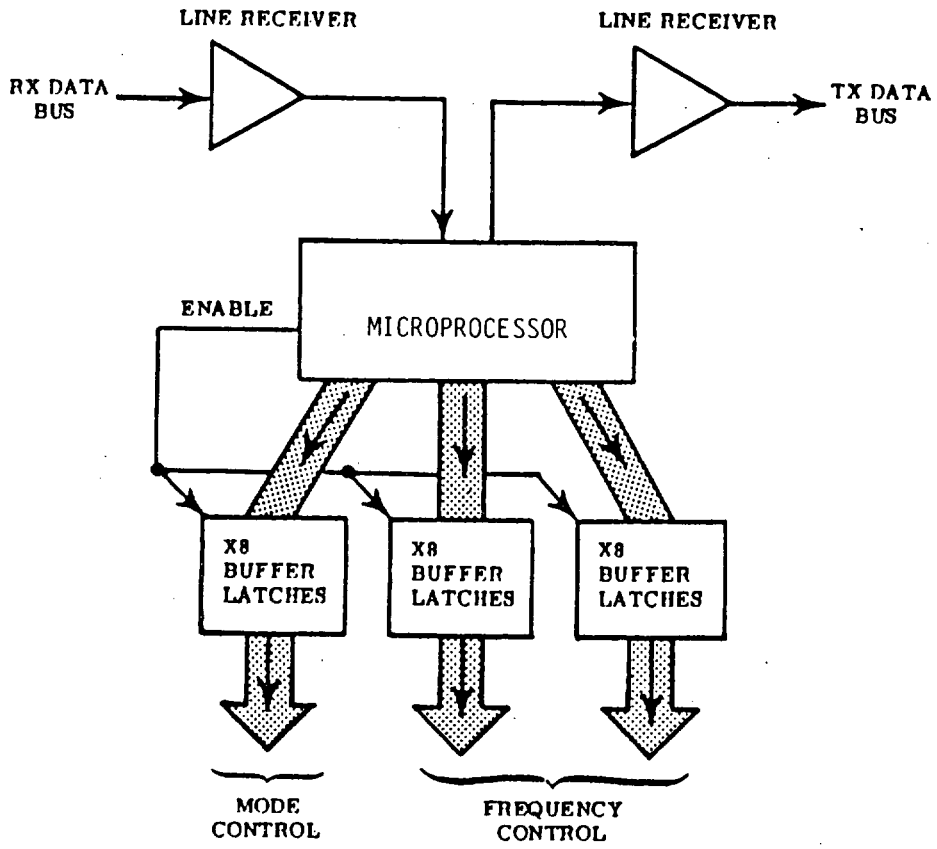


FIGURE 2. AN/WSC-3(V)8 (replaces present preset memory drawer of AN/WSC-3(V)7 version).

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3.2.16.1.1.1 Message format. The message format shall be as specified in 3.2.16.1.1.2 through 3.2.16.1.1.4.

3.2.16.1.1.2 Character description. In accordance with the Standard interface (low level) paragraph of MIL-STD-188, each character transmitted shall consist of 10 elements, each having the same time interval transmitted at 2400 baud. There shall be one zero (space) START element, eight DATA elements (mark or space-NRZ) and one (mark) STOP element. The intercharacter interval shall be 400 microseconds (µs) minimum to any length, maximum, between messages and shall be 0 µs minimum to 1000 µs maximum between characters in a message. The intercharacter interval shall be of the same sense as the STOP element as shown in FIGURE 3.

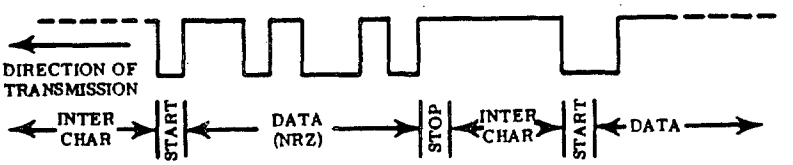
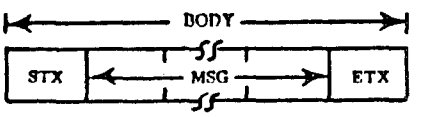


FIGURE 3. Character description.

3.2.16.1.1.3 Character content. The RSC shall have the capability to recognize or to generate the ASCII characters specified in a and b, as a minimum:

Character name	Abbreviation	Equivalent octal code	Equivalent hex code
a. Start of text	STX	002	02
b. End of text	ETX	003	03

Central interactive controller (CIC) messages to remote serial control (RSC)



The body consists of an STX, five 8-bit digital words, and an ETX. The format of the body shall be as specified in TABLE V.

TABLE V. Character content.

Bit No. 1/	Character						
	1(STX)	2(ADDRESS) 2/	3(STATUS)	4(FREQ. 1)	5(FREQ. 2)	6(MODE)	7 (ETX)
2	0	LSB ADD 3/	COMMAND	LSB TENTHS	LSB UNITS	LSB MODE	1
3	1	ADD	STATUS 1	TENTHS	UNITS	MODE	1
4	0	ADD	STATUS 2	TENTHS	UNITS	MODE	0
5	0	ADD	STATUS 3	MSB TENTHS	MSB UNITS	MSB MODE	0

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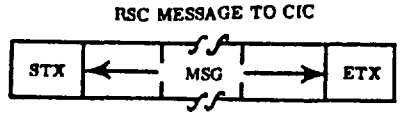
TABLE V. Character content. (continued)

Bit No. 1/	Character						
6	0	$\overline{\text{MSB ADD}}$	$\overline{\text{STATUS 4}}$	$\overline{\text{LSB HUNDREDTHS}}$	$\overline{\text{LSB TENS}}$	1	0
7	0	1	$\overline{\text{STATUS 5}}$	$\overline{\text{MSB HUNDREDTHS}}$	$\overline{\text{TENS}}$	1	0
8	0	1	$\overline{\text{STATUS 6}}$	1	$\overline{\text{TENS}}$	1	0
9	0	1	1	$\overline{\text{HUNDREDS}}$	$\overline{\text{MSB TENS}}$	1	0

1/ Bit No. 1 is the start bit. The LSB shall follow immediately after the start bit.

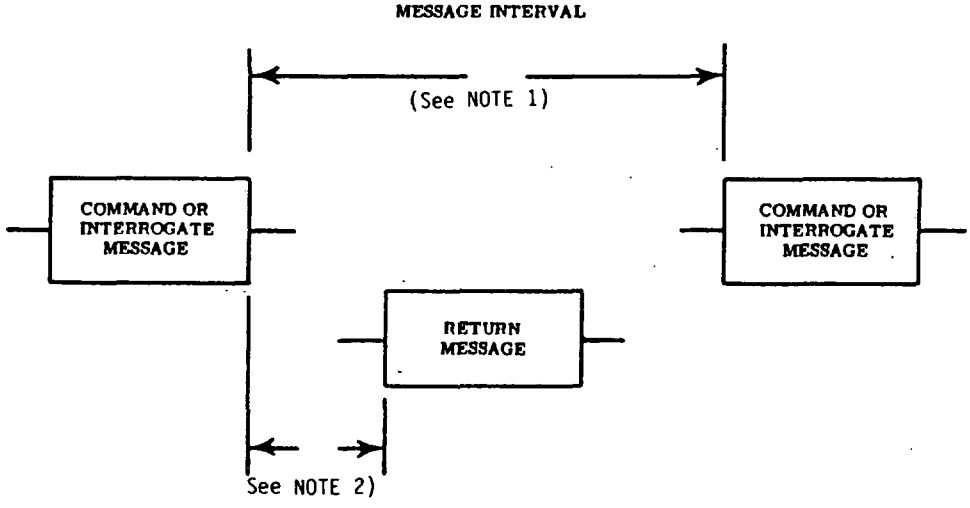
2/ Address is optional. When not used, all bits shall be a logical 1.

3/ $\overline{\text{LSB ADD}}$: Bar means negative logic.



The message is of the same format as the CIC to RSC message.

3.2.16.1.1.4 Message interval. The message interval shall be as specified in FIGURE 4.



NOTES:

1. End of command or interrogate message to beginning of next command or interrogate message is 30 ms nominal.
2. End of command or interrogate message to beginning of return message shall be 0 µs minimum to 1000 µs maximum.
3. Radio status shall be verified 2 times per second nominal when the keyboard is not in use. When keyboard is in use radio status interrogation shall be interrupted briefly.

FIGURE 4. Message interval.

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3.2.16.1.2 Control requirements. The control functions specified in a through c shall be provided:

a. Command and status
b. Frequency
c. Mode

The transmitter key(s) shall be activated by separate remote line(s). Narrowband and wideband operation shall be accomplished by a separate remote line. Manual local control with the exception of 20 channel preset shall be unaffected by the serial control modification.

3.2.16.1.3 Retransmit of data. The controller shall poll the equipment to obtain current operating frequency, thus ensuring that a power outage or malfunction has not occurred affecting the status of the equipment. The equipment shall not remember prior frequency and mode when primary power is shut off. Refresh from the bus controller shall be provided when primary power is reapplied. Design of the RSC drawer shall incorporate software to create invalid frequency information to the synthesizer when power is first applied or restored after an outage. When the synthesizer receives invalid frequency instructions, a malfunction condition shall cause the external alarm contacts to open, thus providing an indication that a power interrupt has occurred.

3.2.17 AN/WSC-3(V)9 configuration. The AN/WSC-3(V)9 version of the AN/WSC-3(V) shall conform to the requirements specified herein for the AN/WSC-3(V)3 except as specified in 3.2.17.1 through 3.2.17.7.

3.2.17.1 5 kHz tuning. The AN/WSC-3(V)9 shall have expanded frequency coverage beyond the 7000 discrete 25-kHz channels specified in 3.2.4 in that the AN/WSC-3(V)9 shall have the added frequency offset capabilities of +5 kHz, +10 kHz, 0 kHz, -5 kHz, and -10 kHz via front panel control.

3.2.17.2 COMM MODE switch. The function of the COMM MODE switch shall be expanded for the AN/WSC-3(V)9 application to include SATCOM OFFSET kHz switch positions of +10, +5, 0, -5, and -10 in addition to the normal LOS and SATCOM 25-kHz switch positions.

3.2.17.3 Modulation. Performance of the AN/WSC-3(V)9 in any of the SATCOM OFFSET kHz, COMM MODE positions shall be applicable only to the FSK modulation mode, which shall be half duplex operation. Other modulation modes may be selected in conjunction with SATCOM OFFSET kHz communication modes, but AN/WSC-3(V)9 performance is unspecified.

3.2.17.4 BER (white gaussian noise). The modem performance-related BER versus E_b/N_o for the AN/WSC-3(V)9 in either the SATCOM 25-kHz, or any of the SATCOM OFFSET kHz communication modes, is relaxed from the BER specified in 3.2.8.7.2.4 and for 75 baud FSK operation is:

E_b/N_o (dB)	BER (maximum)
21.0	1 in 10^5
19.0	1 in 10^3

3.2.17.5 Transmitter spurious radiation. The transmitter spurious radiation for the AN/WSC-3(V)9 shall be as specified in 3.2.6.3, except radio performance is not specified ±200 kHz from the carrier frequency.

3.2.17.6 Receiver spurious response. The receiver spurious radiation for the AN/WSC-3(V)9 shall be as specified in 3.2.7.5, except radio performance is not specified ±200 kHz from the receive center frequency (f_o).

3.2.17.7 Receiver frequency off-set. The receiver frequency off-set capability specified in 3.2.4.1 is expanded for the AN/WSC-3(V)9 configuration to include the specified TABLE III frequency off-sets:

SATELLITE	Receive frequency off-set
FLTSAT COM 5 kHz	73.1 MHz
Servicable defense SAT	72.9 MHz

3.2.18 AN/WSC-3(V)12 configuration. The equipment shall conform to the requirements specified herein for the AN/WSC-3(V)12 and to the additional requirements specified in 3.2.18.1 through 3.2.18.2.

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3.2.18.1 Cooling. The equipment shall be cooled by air drawn in from the right side, referenced facing the front panel. The existing front panel aperture and filter will be replaced by a cover plate over the existing hole. No air filter will be required for the new aperture.

3.2.18.1.1 Air intake aperture design. The air intake aperture shall be designed to maintain the RF interference integrity of the cabinet.

3.2.18.1.2 Air intake aperture dimensions. The air intake aperture shall conform to the dimensional requirements of FIGURE 5.

3.2.18.2 Remote performance monitoring. Remote BITE outputs shall be present on pins of connector J2 to provide GO, NO/GO indications of the functions specified in TABLE VI.

TABLE VI. Remote BITE outputs.

Pin Number	Function	Load	GO voltage	NO/GO
J2-9	Synthesizer	10 kohm	0.33 - 0.67V	< 0.3V
J2-10	Power supply sum	10 kohm	0.33 - 0.67V	< 0.3V or > 0.7V
J2-12	5 MHz standard output	10 kohm	0.33 - 0.67V	< 0.3V
J2-13	5 MHz standard oven	10 kohm	0.33 - 0.67V	< 0.3V
J2-15	44.5 MHz VCXO output	10 kohm	0.33 - 0.67V	< 0.3V
J2-16	44.5 MHz VCXO oven	10 kohm	0.33 - 0.67V	< 0.3V
J2-7	Return for 9, 10, 12, 13, 15, 16			

3.3 Electromagnetic interference characteristics. The equipment shall conform to the requirements of Classes A4 and A5 of MIL-STD-461, except that the CE01 narrowband test limit is relaxed by 7 dB and the CE03 broadband test limit at 20 kHz in the AM transmit mode is relaxed by 3 dB. Where tests required in MIL-STD-461 duplicate those required by this specification, the requirements and limits of this specification shall govern and the duplicate tests shall not be performed. The test limit of Method RE02 of MIL-STD-461 at the carrier frequency is relaxed by 20 dB.

3.4 Identification and marking. Identification and marking shall be in accordance with the Identification and marking paragraph of MIL-E-16400. Items with identification plates located in equipment areas not readily visible shall have additional marking as specified in a through c:

a. Item name
b. Item nomenclature (if applicable)
c. Item serial number (if applicable)

This information shall be legibly and permanently marked by engraving, stamping, stenciling, silk screening, or other suitable means, and shall be visible without removing the item from its associated assembly.

3.5 Safety. Safety criteria requirements of MIL-E-16400 shall be incorporated in the design of the equipment to preclude personnel from accidentally coming into contact with hazardous electrical potentials.

3.5.1 Leakage current. If the equipment leakage current is in excess of 5 milliamperes (mA) a warning plate shall be attached to the front panel reading:

DANGER - DO NOT ENERGIZE OR CONNECT POWER TO THIS EQUIPMENT
UNLESS FRAME AND ALL EXPOSED METAL PARTS ARE GROUNDED

3.6 Quantitative reliability requirements. The equipment, including BITE, shall have a minimum acceptable upper mean-time-between-failures (MTBF) (θ_0 as defined by MIL-STD-781) of 4500 hours, and a minimum acceptable lower MTBF (θ_1 as defined by MIL-STD-781) of 1500 hours.

3.7 Maintainability requirements. Maintainability requirements shall be as specified in 3.7.1 through 3.7.5.2.

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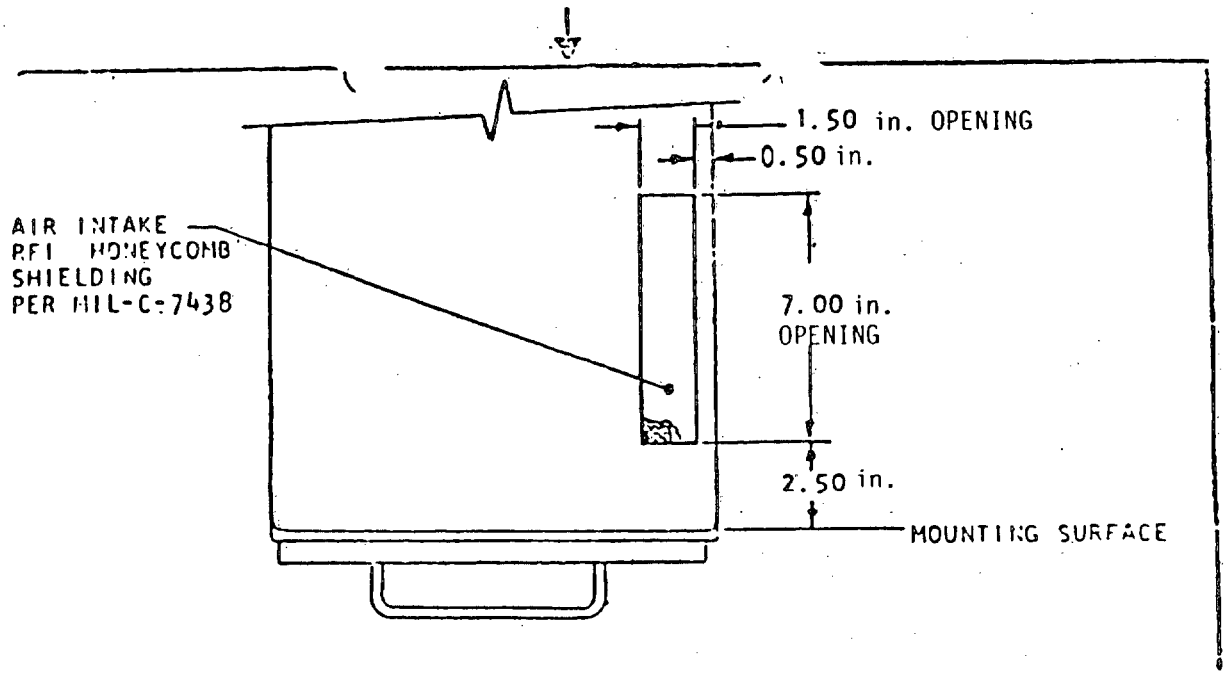


FIGURE 5. Air intake aperture dimensions.

3.7.1 Quantitative maintainability requirement. The equipment shall have an equipment repair time (ERT) of not more than 10 minutes when corrective maintenance by module replacement is performed. The ERT for a failed module or assembly shall be less than 2 hours.

3.7.2 Maintainability design criteria. The maintenance design of the equipment shall permit isolation of a failure in the equipment to a single module or assembly by the use of BITE, which shall be operated from the front panel of the equipment. After the failure is isolated to a functional module or assembly within the equipment, it shall be possible to isolate the failure to a part(s) in the module or assembly by the use of extender cables and general purpose electronic test equipment (GPETE) to the maximum extent practical. All maintenance shall be possible without the necessity for rear or side access. Drawer type construction, including slides with ±90 degree tilt and locking shall be required. The maintainability design requirement shall be to maintain the equipment aboard ships and submarines by a) isolation of the failure to a single module or assembly by the use of BITE, b) replacing the failed module or assembly to restore the equipment to operational readiness, and c) troubleshooting and repairing the module or assembly using extender cables (or cards) and GPETE by replacement of the failed part(s) during the time the equipment is not required for operation. The alignment of the equipment shall be possible with all modules in place with the use of extender cards.

3.7.3 Module extender cables. Module extender cables (or cards) shall be provided for all modules and assemblies of the equipment. Quantities of extender sets shall be as specified in the contract (see 6.2.1). Module extender cables (or cards) for modules associated with SATCOM shall not be delivered with the LOS versions of the AN/WSC-3(V).

3.7.4 Recalibration of the VCXO. Recalibration of the VCXO shall not be required during a 90 day mission. The VCXO may be recalibrated during the initial 30 day installation or replacement period.

3.7.5 Maintenance requirements. Maintenance requirements shall be as specified in 3.7.5.1 and 3.7.5.2.

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3.7.5.1 Maintenance concept. In keeping with the reliability requirements for this equipment, the maintenance concept for repair of failures shall be replacement of plug-in type modules and printed circuit boards (PCBs) to restore the equipment to operating specification. The BITE shall be used to the extent specified herein to localize failures. The complexity of modules and PCBs, subject to removal because of part failures, should be kept at a level such that the repair can be accomplished at the organizational or user level by an electronic technician of skills no greater than those of an ET-3. When the complexity of a repairable item is deemed beyond the capability of an ET-3, the contractor shall recommend the maintenance level capable of restoring the repairable item to its specified performance level.

3.7.5.2 Maintenance level recommendations. Each repairable item shall have the recommendation of the contractor for the Navy maintenance level capable of effecting the required maintenance. The NAVPERS 130689 manual shall be used to determine the maintenance level assignment of each maintenance task. Ships and submarines shall be considered to possess only the repair capabilities of the organizational maintenance level as defined in 6.5.1. Recommendations requiring the capabilities and resources of intermediate and depot maintenance levels shall have the requirement and maintenance task fully described and documented.

3.8 Workmanship. Workmanship shall be as specified in 3.8.1 and 3.8.2.

3.8.1 General workmanship. Workmanship shall conform to the Workmanship paragraph of MIL-E-16400. Soldering shall conform to the requirements of WS-6536D.

3.8.2 Workmanship screen. All equipment shall withstand a defect detection vibration screen of random type vibration at $0.004g^2/Hz$ ±3dB from 80 to 350 Hz and temperature cycling as specified in 4.5.1.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (Group A) (see 4.4.1)
    2. Production control inspection (Group B) (see 4.4.2)
    3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection of preparation for delivery (see 4.9)

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4.3 First article inspection. Unless otherwise specified (see 6.2.1), two units of a given configuration (SATCOM or LOS) shall be submitted for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE V.

4.4 Quality conformance inspection. Quality conformance inspections shall be as specified in 4.4.1 through 4.4.3.

TABLE VII. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination	3.1	4.5.1	X[1/]		X	
Weight	3.1.8	4.5.1	X		X	
Size	3.1.8	4.5.1	X		X	
Parts and materials	3.1, 3.1.5.1, 3.2.11.3	4.5.1	X[1/]	X[1/]		
Finish	3.1	4.5.1	X[1/]	X[1/]		
Marking	3.4	4.5.1	X[1/]	X[1/]		
Safety	3.5	4.5.1	X[1/]	X[1/]		
Workmanship	3.8	4.5.1	X[1/]	X[1/]		
Pre- and post-test functional verification		4.5.2	X[1/]		X[1/]	X[1/]
Shock	3.1.2	4.5.1	X[1/]			
Vibration	3.1.2	4.5.1	X[1/]			X[1/]
Inclination	3.1.2.1	4.5.1	X[1/]			X
Temperature	3.1.3	4.5.1	X[1/]			X
Humidity	3.1.3	4.5.1	X[1/]			X
Airborne and structureborne noise	3.1.4	4.5.1	X			
Equipment cooling	3.1.7	4.5.1	X		X	
Enclosure	3.1.9	4.5.1	X[1/]			X[1/]
Altitude	3.1.10	4.5.3	X			
Nuclear effects	3.1.20	4.5.5	X			
Reliability	3.6	4.6	X			
Maintainability	3.7	4.7	X			
Input power	3.2.1	4.5.1	X		X	
T/R turn around time	3.2.2	4.5.1	X		X	
Emissions	3.2.3	4.5.1	X	X	X	
Frequency coverage	3.2.4	4.5.1	X		X	
Channel selection	3.2.5	4.5.1	X		X	
RF power output	3.2.6.1	4.5.1	X	X		
Mismatched load	3.2.6.2	4.5.1	X		X	
Spurious radiation	3.2.6.3	4.5.1	X		X	
Harmonic attenuations (transmit)	3.2.6.4	4.5.1	X		X	
AM harmonic distortion (transmit)	3.2.6.5	4.5.1	X	X		
Intermodulation (transmit)	3.2.6.6	4.5.1	X		X	
FM noise (transmit)	3.2.6.7	4.5.1	X	X		
Secure voice interface (transmit)	3.2.6.8 through 3.2.6.8.3	4.5.1	X		X	
Plain text interface (transmit)	3.2.6.9 through 3.2.6.9.2	4.5.1	X		X	

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TABLE VII. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
LINK 11 (transmit)	3.2.6.10 through 3.2.6.10.1.4	4.5.1	X		X	
Selectivity	3.2.7.1	4.5.1	X		X	
Sensitivity	3.2.7.2	4.5.1	X	X		
Noise figure	3.2.7.3	4.5.1	X		X	
Overload	3.2.7.4	4.5.1	X		X	
Spurious response	3.2.7.5	4.5.1	X		X	
Image rejection	3.2.7.6	4.5.1	X		X	
IF rejection	3.2.7.7	4.5.1	X		X	
AM harmonic distortion (receiver)	3.2.7.8	4.5.1	X	X		
Intermodulation (receiver)	3.2.7.9	4.5.1	X		X	
FM noise (receiver)	3.2.7.10	4.5.1	X	X		
Automatic gain control (AGC)	3.2.7.11	4.5.1	X	X		
Pulse blanking	3.2.7.12	4.5.1	X		X	
Squelch and signal presence	3.2.7.13 through 3.2.7.13.3	4.5.1	X	X		
Secure voice interface (receiver)	3.2.7.14 through 3.2.7.14.3	4.5.1	X		X	
Plain text interface (receiver)	3.2.7.15 through 3.2.7.15.2.1.3	4.5.1	X		X	
LINK 11 (receiver)	3.2.7.16 through 3.2.7.16.1.3	4.5.1	X		X	
Handset, voice operation	3.2.8.3	4.5.1	X	X		
Audio output	3.2.8.4	4.5.1	X	X		
AM	3.2.8.5 through 3.2.8.5.4.2	4.5.1	X		X	
FM	3.2.8.6 through 3.2.8.6.5	4.5.1	X		X	
Digital data	3.2.8.7 through 3.2.8.7.3.4	4.5.1	X		X	
75 BPS PSK data modem	3.2.8.8 through 3.2.8.8.3	4.5.1	X		X	
300, 1200, 2400, 4800, 9600 BPS PSK modem	3.2.8.9 through 3.2.8.9.4	4.5.1	X		X	
External modem interface	3.2.9	4.5.1	X		X	
Antenna group interface	3.2.10	4.5.1	X		X	
Local and remote operation	3.2.11 through 3.2.11.4	4.5.1	X [1]	X [1]		
LINK 11 select	3.2.12	4.5.1	X		X	
AM diphase and FM baseband control	3.2.13	4.5.1	X		X	
Secure and plain text control	3.2.14	4.5.1	X		X	
BER FM secure text mode	3.2.15	4.5.1	X		X	
Electromagnetic interference	3.3	4.5.11	X			
Workmanship screen	3.8.2	4.8.1.2		X		
Leakage current	3.5.1	4.5.7	X		X	

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TABLE VII. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
System interface	3.1.1	4.5.4	X			
Serial control	3.2.16.1	4.5.10	X		X	
SATCOM offset tuning	3.2.17.1	4.5.8		X		
Alternate intake and remote BITE	3.2.18	4.5.9		X		

1/ These tests shall be performed in conjunction with: C9351/WSC-3 Control Indicator

4.4.1 Production inspection (Group A). Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. The SATCOM and LOS AN/WSC-3(V) radios shall be subjected, as a minimum, to all production tests specified in TABLE VII, Group A, including subparagraphs. The BITE controls shall be activated. Any NO-GO readings shall constitute an equipment failure.

4.4.2 Production control inspection (Group B). Production control inspection including sampling, shall conform, as a minimum, to TABLE V, Group B and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection is to be conducted on combined quantities of SATCOM and LOS versions. Samples for each version shall be based on the ratio of the quantities of each version to the total (combined) lot size. Samples shall be chosen from equipments which have passed production inspection. The equipment shall satisfactorily conform to the requirements of production control inspection prior to release for shipment.

4.4.2.1 Acceptable quality level. Acceptable quality level for each inspection shall be 6.5 percent.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot, using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be performed on units which have been subjected to and have passed production inspection. Environmental testing shall consist, as a minimum, of the tests of TABLE VII, Group C. Quantities of AN/WSC-3(V) SATCOM versions and LOS versions shall not be mixed together, but shall be considered separately for sampling purposes.

4.4.3.1 Environmental inspection samples. Samples for environmental inspection shall be selected without regard to their quality in accordance with a and b:

a. One sample from the first 50 units fabricated
b. One sample from each successive 100 units, or fraction thereof, produced

4.4.4 Nonconformance corrective action and retest. If a sample unit fails the inspection specified in 4.4.2 or 4.4.3, the contractor shall immediately investigate the cause of failure and shall implement the corrective action necessary to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. The equipment shall then be retested.

4.4.5 Reinspection of production control and environmental test units. Unless otherwise specified in the contract (see 6.2.1), sample units which have been subjected to and have passed production control and environmental inspections may be accepted on the contract provided they are resubjected to, and pass, production inspection specified in 4.4.1 after repair of all visible damage.

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4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.11.

4.5.1 Tests and examinations. Tests and examinations specified herein shall be conducted in accordance with the applicable inspection procedures of MIL-E-16400.

4.5.2 Pre- and post-test functional verification. Prior to proceeding with any of the Group B or C test methods, the test item shall be operated under the standard ambient conditions and a record made of all data necessary to determine compliance with required performance. These data shall provide the criteria for checking satisfactory performance of the test item either during or at the conclusion of the test, or both, as required. When operation of the test item is required during the test exposure, the operation and performance checks shall be of sufficient duration or shall be repeated at appropriate intervals, to ensure a record of comprehensive comparative data for comparison with data recorded under standard ambient conditions. The test item shall be visually examined and a record made of any damage or deterioration resulting from the test. If a test chamber is used for the test, a visual examination of the test item within the chamber shall be performed at test conditions, when possible. Upon completion of the test, a visual examination of the test item shall be performed again, after the test item has been returned to standard ambient conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from conforming to operational service or maintenance requirements, shall provide reason to consider the test item as having failed to withstand the conditions of the test. Leakage or degradation of impregnating compounds shall be considered damage and cause for rejection.

4.5.3 Altitude test. The altitude test shall be conducted in accordance with MIL-STD-810, Method 500.2, Procedure I, to demonstrate conformance with 3.1.10.

4.5.4 System interface testing. The contractor shall perform integration tests to demonstrate the successful interface of the contractor-furnished equipment with the equipment specified in 3.1.1, 3.2.8.3, and 3.2.10.

4.5.5 Nuclear effects tests. Nuclear effects tests shall be performed to demonstrate compliance with 3.1.20.

4.5.6 Controls and control circuit tests. The equipment shall be tested to determine the suitability of controls and control circuits for satisfactory mechanical and electrical operation in accordance with 3.1.6 and 3.2.11.

4.5.7 Leakage current test. The leakage current test shall be as specified in 4.5.7.1 and 4.5.7.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.7.1 Equipment connections. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened during the test. The equipment shall be connected as shown in FIGURE 6, if it is connected to single-phase or DC source power.

4.5.7.2 Measurement. Leakage current shall be measured on equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage shall be measured across the 1500-ohm resistor at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. A measurement of 7.5 V as determined by the voltage-drop method shall correspond to a current of 5 mA. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in the test diagram. The open safety ground conductor shall be reconnected immediately after the test is completed.

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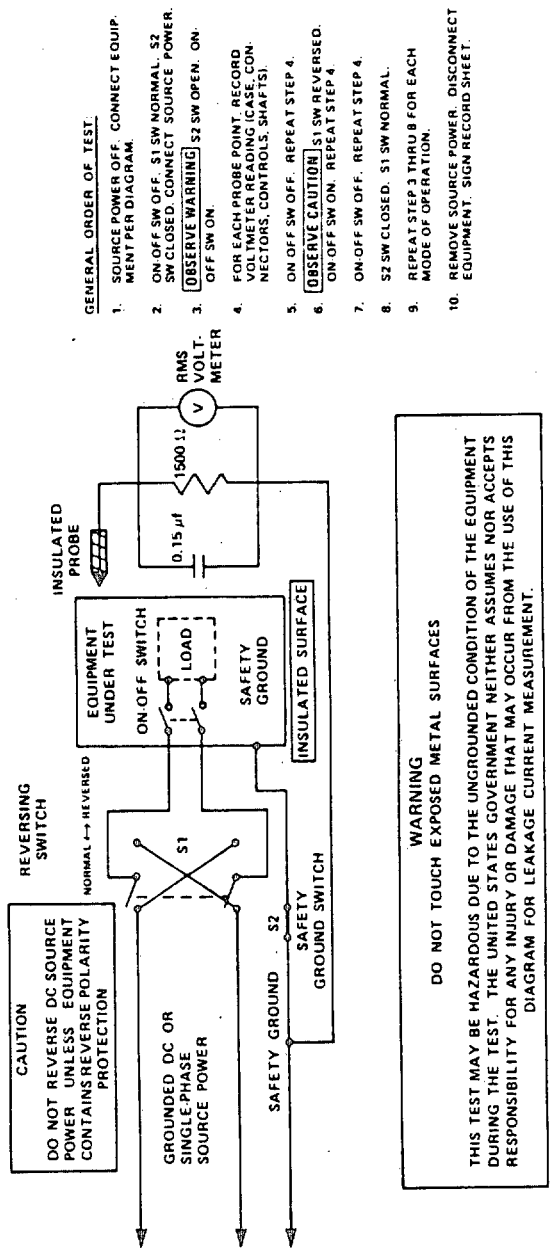


FIGURE 6. Single-phase or DC test diagram for leakage current measurement.

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4.5.8 SATCOM offset tuning. The (V)9 equipments shall be tested to demonstrate conformance with the ±5 kHz and ±10 kHz offset tuning of 3.2.17.

4.5.9 Alternate intake and remote BITE. The (V)12 equipments shall be examined and tested for compliance with the alternate air intake and the remote BITE requirements of 3.2.18.

4.5.10 Serial control tests. Testing shall be conducted to ensure compliance with the requirements of 3.2.16. A Government furnished equipment (GFE) commercial quality CIC (see 6.4) may be utilized as an aid in the contractor's in-plant evaluation of a complete equipment system. However, final evaluation shall be conducted utilizing the actual Coast Guard controller at a Coast Guard installation. FIGURE 7 illustrates the components of an evaluation system using a simulated controller.

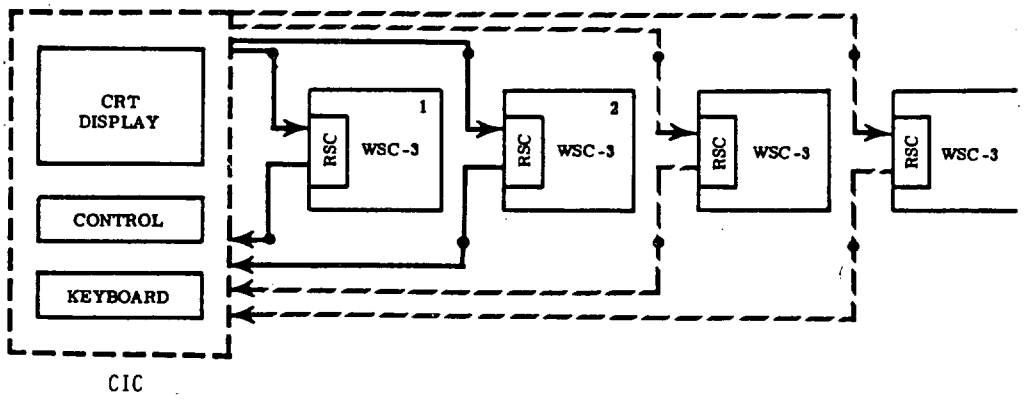


FIGURE 7. Simulated control system (two or more radios used).

4.5.11 Electromagnetic interference tests. Electromagnetic interference emission, susceptibility, and self-compatibility tests conducted for first article inspection shall be performed in accordance with MIL-STD-462.

4.6 Reliability. Reliability tests shall be as specified in 4.6.1 through 4.6.4.

4.6.1 Reliability testing. The reliability of the equipment shall be demonstrated in accordance with MIL-STD-781. The AN/WSC-3(V) shall be qualified with respect to reliability using MIL-STD-781, Test plan XVC. Test conditions of MIL-STD-781, APPENDIX B, Category 3A, shall apply with the exceptions that the ambient temperature shall be a constant 50°C and that no on/off cycling, or input voltage cycling is required. The equipment shall be operated in a 50 percent transmit/ 50 percent receive duty cycle every 6-hour time period (3 hours transmit/3 hours receive), during the test. Testing shall proceed to an accept or reject decision in accordance with the criteria specified for Test plan XVC in the Summary of test plans (fixed length) table of MIL-STD-781.

4.6.2 Monitored parameters. The parameters specified in a through d shall be measured at least once every 12 hours on each equipment throughout the entire reliability test:

a. RF power output (see 3.2.6.1)
b. Receiver sensitivity (see 3.2.7.2)
c. Frequency stability (see 3.2.4.3)
d. Modem performance (see 3.2.8)

4.6.3 Failures. A failure occurs any time the equipment fails to conform to any of the specified requirements. The relevancy or nonrelevancy of failures shall be as defined in the Failure categories paragraph of MIL-STD-781.

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4.6.3.1 Adjustments. Adjustments required during the reliability test, except for external operator controls, shall be considered relevant failures.

4.6.4 Disposition of tested equipment. Upon successful completion of the reliability test, the equipment shall be refurbished, as necessary, and submitted for production acceptance tests. The equipment may be delivered as part of the contract quantity provided it conforms to all requirements of the contract. If the equipment fails the reliability tests, the action of the Reject in reliability qualification phase paragraph of MIL-STD-781 shall be implemented.

4.7 Maintainability and BITE demonstration. The contractor shall perform a maintainability demonstration on the AN/WSC-3(V) radio at the organizational level when specified in the contract. The demonstration shall be performed by a technician with the experience equivalent to that of an electronics technician third class, and shall be used to verify that the equipment conforms to ERT requirements of 3.7.1.

4.7.1 Demonstration conditions. Demonstration conditions shall be as specified in 4.7.1.1 and 4.7.1.2.

4.7.1.1 Maintainability. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 (accept or reject criteria, Method 4) by replacement of subassemblies (modules, individual printed circuit cards) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. The contractor shall determine 60 candidate faults for the demonstration. A representative of the procuring activity shall select a sample of 20 faults from these candidate faults to be used in this part of the demonstration. For fault simulation, a number shall be assigned to each component or part within a subassembly or chassis-mounted component or part. Fault simulation for modules shall be accomplished by introduction of faulty parts corresponding to the number selected.

4.7.1.2 Demonstration environment. The demonstration at the organizational level shall be performed in a space and temperature environment similar to that in which the equipment will be installed.

4.7.2 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the equipment maintainability demonstration.

4.7.3 Technical documentation. Technical documentation to be used in the demonstration shall be limited to the technical manual.

4.7.4 Rejection. Failure to conform to any of these requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified.

4.7.5 Acceptance. The maintainability demonstration shall exhibit that the equipment has conformed to the accept criteria prior to the delivery of any equipment.

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 and 4.8.2.

4.8.1 General workmanship. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirement of 3.8.1.

4.8.1.2 Workmanship screen. Vibration (see 4.8.2.1) and temperature cycling (see 4.8.2.2) shall be performed on each equipment. Vibration shall be performed prior to temperature cycling. The vibration may be performed at the module, drawer, or end item level. All the hardware including cables and connectors shall be exposed to vibration.

4.8.1.2.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 8. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz, and 100 Hz, for vibration frequencies above 500 Hz, shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the rms acceleration level. The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

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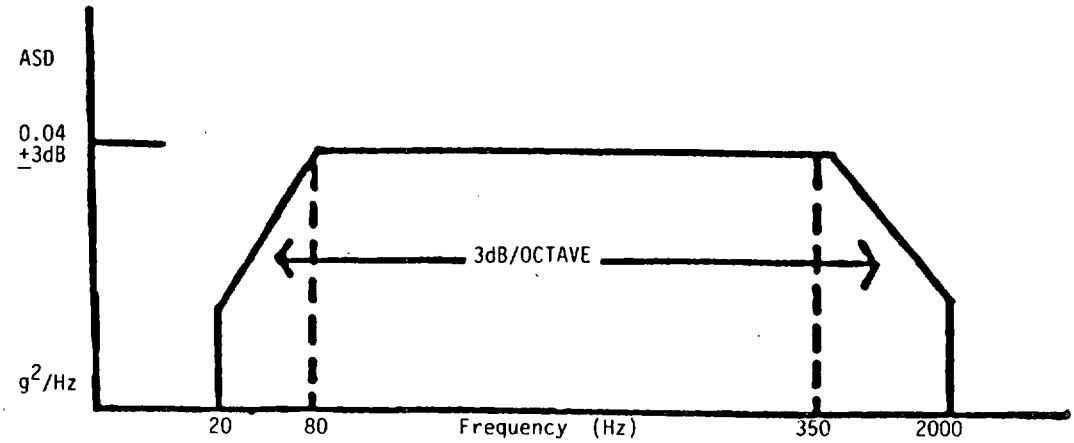


FIGURE 8. Random vibration curve.

4.8.1.2.2 Temperature cycling. Each unit shall be subjected to 10 cycles of the temperature curve shown in FIGURE 9. The temperature rate of change shall be not less than 5°C per minute. Unit power shall be turned on and off at the indicated times. The unit shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the unit reaches 80 percent of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cumulative total of 10 cycles.

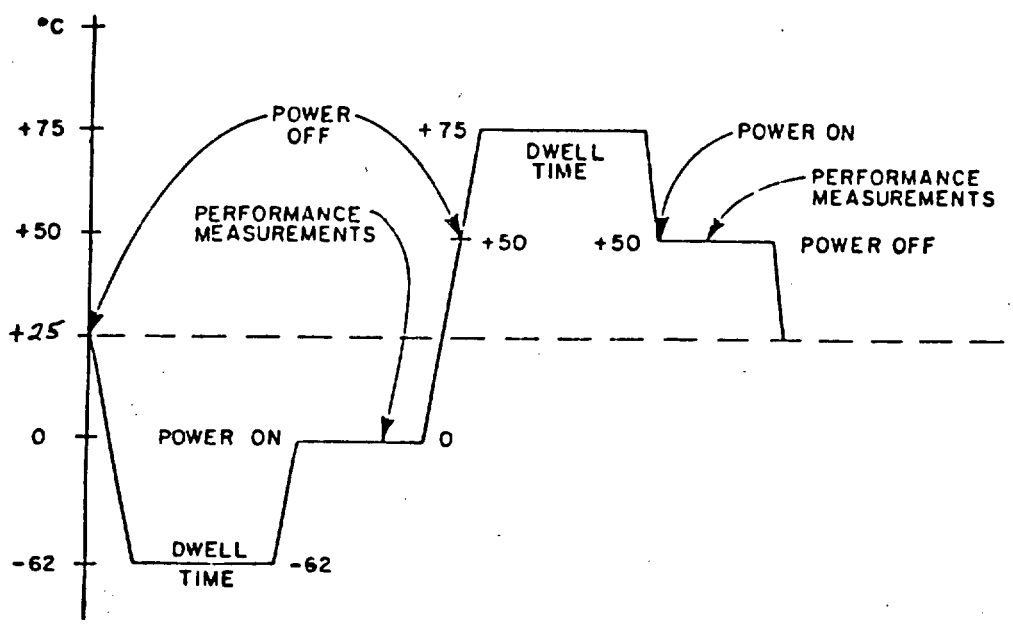


FIGURE 9. One cycle of temperature curve (not to scale).

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4.8.1.2.2.1 Equipment operation during temperature cycling. The SATCOM radio sets shall be operated in the PSK data mode, key down, delivering rated power for 50 percent of the operating time with the other 50 percent of the operating time in the PSK data receive mode. The LOS radio sets shall be operated in the AM mode, key down, delivering rated power for 50 percent of the operating time, with the other 50 percent of the operating time in the receive mode. The test duty cycle shall be 2 hours for the transmitter and 2 hours for the receiver. Twice every 24-hour period the BITE shall be exercised through all positions to detect any component failures.

4.9 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The UHF satellite and LOS communications sets are for installation in ships and submarines. The equipments shall be capable of and shall be used for transmitting and receiving information over the frequency range of 225.000 MHz to 399.975 MHz. The equipment shall be compatible for operation with the associated equipment specified herein.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify:

a. Title, number, and date of this specification
b. When modules associated with the SATCOM version are to be delivered with the LOS version (see 1.4)
c. Quantity of module extender sets (see 3.7.3)
d. Quantity of C-9351/WSC-3 remote units for LOS radios (see 3.2.11.3)
e. Number of first article samples required if other than specified in 4.3
f. When reinspected production control and environmental test units may not be acepted (see 4.4.5)
g. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List, incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
3.5	Safety Statement	UDI-H-20442
3.4	Request, Government Nomenclature Nameplate Approval, Serial Numbers	UDI-P-22502B
3.6	Plan, Reliability Program	DI-R-2113
3.7	Plan, Maintainability Program	DI-R-2127
3.7.5.2	Recommendations, Maintenance Level	UDI-L-22340
4.1.2	Quality Program Plan	DI-R-5297A
4.3	First Article Inspection Report	DI-T-4902
4.4	Test Plans	DI-T-3709A
4.5.1	Test Procedures	UDI-T-22710A
4.7.1	Plan, Maintainability Demonstration	DI-R-2129
4.7.1	Plan, Maintainability Demonstration	DI-R-2130A
4.6	Procedures, Reliability Tests	DI-R-7035

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(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) should be a first production item. The first article should consist of two unit(s) (see 4.3). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 CIC. The CIC is constructed from commercially available systems components such as Hewlett-Packard 2644 display and keyboard terminal, and a commercially available microprocessor system, programmed to conform to the particular system requirements. This central controller communicates with the equipments on data lines as shown in the system block diagram in FIGURE 7. The operator can then modify the frequency or mode of that link through the keyboard. Frequency can be controlled by inputting either the preset channel number or frequency. Preset frequency information will be contained on magnetic tape in the CIC. The preset frequency information can be changed from the keyboard.

6.5 Definitions. Definitions of terms used in this specification are given in 6.5.1 through 6.5.3.

6.5.1 Organizational maintenance. Organizational maintenance is defined as that maintenance which is the responsibility of and performed by using organizations on any assigned equipment. The phases normally consist of inspection, servicing, lubricating, adjusting and replacing of parts, minor assemblies and subassemblies. For the Navy, this is normally shipboard maintenance of Navy equipment, including scheduled preventive maintenance. In addition, the complexity of tasks assigned to this maintenance level should not exceed the capabilities of the required specialty at skill level E-3 or E-4.

6.5.2 Depot maintenance. Depot maintenance is defined as that maintenance performed on material requiring major overhaul or a complete rebuild of parts, assemblies, subassemblies, and end items, including the manufacture of parts, modification, testing and reclamation as required. Depot maintenance serves to support lower categories of maintenance by providing technical assistance and performing that maintenance beyond their responsibility. Depot maintenance provides stocks of serviceable equipment by using more extensive facilities for repair than are available in lower level maintenance activities. For the Navy, this is normally that maintenance performed by the Naval air rework facilities, depot field teams, Naval ammunition depots, Naval ordnance stations, Naval weapons stations, torpedo stations, Naval construction battalion centers and POLARIS missile facilities. This also includes contractor depot level rework activities and at commercial facilities or Navy shipyards (including repair facilities) during availabilities, designated voyage repairs, restricted, technical, regular overhauls and the like. All tasks of complexities exceeding the capabilities of the required specialty at skill levels higher than those stated shall be assigned to this maintenance level.

6.5.3 Intermediate maintenance. Intermediate maintenance is defined as that maintenance which is the responsibility of and performed by designated maintenance activities for direct support of using organizations. The phases normally consist of calibration, repair, or replacement of damaged or unserviceable parts, components, or assemblies; the emergency manufacture of nonavailable parts; and providing technical assistance to using organizations. For the Navy, this is normally maintenance performed by submarine tender ships, and shore activities officially designated as such. Tasks assigned to this maintenance level should not exceed the capabilities of the required specialty at skill level E-4 or E-5.

6.6 GFE. The contracting officer should arrange to furnish the GFE listed in 4.5.10.

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6.7 International standardization agreement. Certain provisions of this specification (identified by paragraph number or similar manner, if appropriate) are the subject of international standardization agreement (insert the ABCA or ASCC standard number, the NATO, STANAG, NETR or NEPR number, or other appropriate documentary reference). When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY-EC

(Project No. 5820-N814(EC))

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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